Master's Thesis

School of Mathematics

On the Sum of the Three Largest Adjacency Eigenvalues of a Graph

A joint moment bound, the structure of extremal graphons, and numerical verification

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4 November 2026

This thesis expands the preprint *A joint moment bound for the three largest graph eigenvalues* (H. Ji, 3 November 2026). All constants quoted here were recomputed in exact arithmetic; the labels *proved*, *verified*, *numerical* and *gap* are used throughout to separate the four kinds of statement.

Declaration

The work presented in this thesis is my own, carried out between August and November 2026. Where a result, an idea, a construction or a constant is due to another author, it is cited at the point of use. Two instances deserve to be singled out here because they concern priority rather than technique.

- The lower-bound constant $c_{\mathrm{w}} = (9 + \sqrt{5} + 2\sqrt{35})/18$ and the optimising weight of the $K_1 \vee C_5$ construction were announced earlier, in conditional form, in the self-archived Zenodo deposits of Newton [13, 14]. No priority is claimed for that constant; see Remark 3.3.
- The observation that the Ky Fan transfer argument needs only a zero diagonal and off-diagonal entries in $[0, 1]$ is already implicit in the proof of [16, Proposition 6.1] and in the matrix setting of [12]. Proposition 6.8 records the resulting weighted form of the main bound but claims no novelty for it.

Every displayed identity, factorisation and positivity assertion reproduced from the preprint was re-checked in exact rational arithmetic and in the quadratic fields $\mathbb{Q}(\sqrt{3})$ and $\mathbb{Q}(\sqrt{5})$, never in floating point. Numerical experiments are reported as numerical evidence only and are never used as a step in a proof.

Haoyang Ji 4 November 2026

Abstract

Let $\lambda_1(G) \geq \lambda_2(G) \geq \cdots$ be the adjacency eigenvalues of a finite simple graph G and write $\Lambda_3(G) = \lambda_1(G) + \lambda_2(G) + \lambda_3(G)$. This thesis proves

$$\Lambda_3(G) \;\leq\; c\,n - 3 \qquad (n \geq 3),$$

where

$$c = \frac{106106902139519}{107310156800000} + \frac{41\sqrt{3}}{240} - \frac{53}{9000000} = 1.284673255723918505\ldots,$$

which improves the coefficient $(3+\sqrt{5})/4 = 1.309016994\ldots$ obtained by Sivashankar, Tang and Wakhare from the rank-three absolute projection constant $\gamma(3) = (1+\sqrt{5})/2$. The constant term -3 is unchanged.

The improvement comes from replacing two separate estimates by one joint estimate. Writing $A = \mathbb{E}|X \cdot X'|$ and $B = |\mathbb{E}X|^2$ for an isotropic law $\mathbb{E}XX^T = I_3$ on $\mathbb{R}^3$ with independent copy X', the classical bounds $A \leq (1+\sqrt{5})/2$ and $B \leq 1$ give exactly the coefficient $(3+\sqrt{5})/4$. We prove instead $A + B \leq 2c$, by an exact identity that couples A and B to the mean deficit, the radial variance, the regression error and the weighted harmonic moments of degrees three and four through a signed quartic majorant. All error terms are retained in the identity, so the certificate is quantitative rather than asymptotic.

Structurally, we prove that the canonical eigenfunction law of every maximising graphon for the variational quantity $\mathcal{M} = \sup_W \Sigma_3(W)$ has at most 24 support points, that the maximiser is almost everywhere a binary step kernel determined by the signs of the dot products of those points, and that every positive clique in the support consists of at most six linearly independent types. The argument combines a second-variation inequality on the actual support, propagation of moment relations, a dimension descent over sign regions, and a separating family of binary observations.

The bound is shown to hold verbatim for $[0,1]$-weighted loopless graphs, and weights are proved not to improve the asymptotic constant. Extensive numerical verification is reported: all 2 131 016 labelled graphs of orders $3 \leq n \leq 7$ satisfy the bound, the unique extremal graph at $n = 6$ is $K_1 \vee C_5$, and blow-ups of $K_1 \vee C_5$ attain $c_{\mathrm{w}}n - 3 + o(1)$ with $c_{\mathrm{w}} = 1.281568196872168\ldots$.

Neither the optimal coefficient nor the extremal graphs are determined. The residual gap $c - c_{\mathrm{w}} = 0.003105058851750628\ldots$ and the structure of maximisers are left open and are discussed in the final chapter.

Keywords. Adjacency eigenvalues; spectral sums; Ky Fan maximum principle; graphons; absolute projection constants; isotropic measures; extremal graph theory.

Acknowledgements

I thank my supervisor for the patience and the many sharp questions that shaped this work, and the authors of [16, 8, 9, 15, 7] for making their preprints available so quickly, which allowed the present argument to be positioned honestly against the current state of the literature. I am also grateful to the authors of the recent preprints on spectral sums for making their work publicly available so promptly, which allowed the present argument to be positioned honestly against the current literature.

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List of Notation

Symbol	Meaning
G, n, A_G	a finite simple undirected graph, its order, its adjacency matrix
$\lambda_1 \geq \cdots \geq \lambda_n$	adjacency eigenvalues in algebraic order, with multiplicity
$\Lambda_k(G)$	$\lambda_1(G) + \cdots + \lambda_k(G)$, signs retained
$M_k(n)$, c_k, τ_k	$\max \Lambda_k$ at order n; $\sup \lambda_k/n$; $\limsup M_k(n)/n$
$K_1 \vee C_5$	the join of a single vertex with a 5-cycle (the wheel W_5)
$H[q]$	equal blow-up of H: each vertex becomes a clique of size q
W, T_W, $\Sigma_k(W)$	a graphon; its integral operator; sum of its first k positive eigenvalues
$\mathcal{M}$	$\sup_W \Sigma_3(W)$
$\|\cdot\|_\square$, $\delta_\square$	cut norm; cut distance between graphons
L, F, ν	$\mathrm{diag}(\lambda_1, \lambda_2, \lambda_3)$ of a maximiser; its canonical frame; the law $F_\# dx$
S, p_i, v_i, B	canonical support; atom masses; support points; $B_{ij} = \mathbf{1}_{\{v_i \cdot v_j > 0\}}$
X, X', $\mathbb{E}XX^T = I_3$	an isotropic law on $\mathbb{R}^3$ and an independent copy
R, U, C, V	radius $\|X\|$; direction X/R; $\mathbb{E}R$; $3 - C^2$
$\mathbf{m}$, B, e	mean $\mathbb{E}X$; $\|\mathbb{E}X\|^2$; $1 - B$
A, D	$\mathbb{E}\|X \cdot X'\|$; the defect $(3+\sqrt{5})/2 - A - B$
M, Z, J, ξ	$\mathbb{E}[RUU^T]$; its traceless part; $1 - \mathrm{tr}\, M^2$; the regression vector $U - MX$
H, K, T_3, T_4	harmonic moment norms $\|T_3\|_F$, $\|T_4\|_F$
g_3, g_4, Ψ_3, Ψ_4	the harmonic Gram kernels of (2.12) and their generating maps
f, σ, h_0	the signed quartic majorant; its slack $f - \|w\| - w$; the classical absolute-value majorant
P, Z_*, $\mathcal{C}$	the slack moment $\mathbb{E}[RR'\sigma]$; the contact moment; its rescaled form
d_0, d_1, d	the three successive defect constants, $d = (3+\sqrt{5})/2 - 2c$
r_{reg}, r_{rad}, r_{tr}, r_{ct}, r_{off}	the retained nonnegative remainders
$\gamma(r)$, Γ_r	the absolute projection constant; the isotropic supremum (2.10)
c_{w}, c	the lower constant $(9+\sqrt{5}+2\sqrt{35})/18$; the proved coefficient

Chapter 1

Introduction

1.1 The spectral sum problem

Throughout this thesis a *graph* means a finite, simple, undirected graph without loops or multiple edges. For a graph G of order n we write

$$\lambda_1(G) \geq \lambda_2(G) \geq \cdots \geq \lambda_n(G)$$

for the eigenvalues of its adjacency matrix A_G, listed in algebraically decreasing order and repeated according to multiplicity. For an integer k with $1 \leq k \leq n$ put

$$\Lambda_k(G) = \lambda_1(G) + \cdots + \lambda_k(G), \qquad M_k(n) = \max\{\Lambda_k(G) : |V(G)| = n\},$$

and, following Nikiforov [26],

$$c_k = \sup\Big\{ \frac{\lambda_k(G)}{|V(G)|} : |V(G)| \geq k \Big\}, \qquad \tau_k = \limsup_{n\to\infty} \frac{M_k(n)}{n}.$$

The signs in Λ_k are retained: for $k \geq 2$ the summands may be negative, and Λ_k is therefore not a norm. Two classical facts make the extremal problem non-trivial and keep it anchored to n. First, $\sum_i \lambda_i(G) = \operatorname{tr} A_G = 0$ and $\sum_i \lambda_i(G)^2 = 2|E(G)|$, so Λ_k is bounded above by $\sqrt{k \cdot 2|E(G)|} \leq \sqrt{k}\, n$ through Cauchy–Schwarz; second, $\Lambda_n(G) = 0$ identically. The interesting range is thus $2 \leq k \ll n$, where the question posed by Hong [21] asks for the best possible order-linear coefficient.

The case $k = 1$ is the spectral radius problem and is classical: $\lambda_1(G) \leq n-1$ with equality only for K_n. The case $k = 2$ was settled only recently [8, 7]. The case $k = 3$ is the subject of this thesis. Sums of this type also control the Ky Fan k-norms $\sum_{i\leq k} \lambda_i^*$ of the singular values, and hence generalise graph energy [26], which explains why the problem has attracted attention from both the graph-theoretic and the Banach-space sides.

1.2 Earlier bounds and projection constants

The first order-linear bounds for individual eigenvalues are due to Hong [20]: for $2 \le k \le n$,

$$\lambda_k(G) \le \frac{n-k}{2}, \tag{1.1}$$

so that $\lambda_2(G) \le n/2 - 1$, an estimate that is sharp for even n by $K_{n/2} \sqcup K_{n/2}$. Hong then asked [21] for the best possible bounds of this shape. By (1.1) and the example $K_{n/2} \sqcup K_{n/2}$ one has $c_2 = 1/2$. Nikiforov [26] proved that c_k and τ_k are finite for every fixed k and obtained bounds for c_k for all $k > 2$, based in part on Taylor's strongly regular graphs and on Kharaghani's construction of Hadamard matrices; he also conjectured [26, Conjecture 4.2] that for every $k \ge 2$ there is an $\varepsilon_k > 0$ with

$$\tau_k \le \frac{1+\sqrt{k}}{2} - \varepsilon_k. \tag{1.2}$$

The reference value $\frac{1}{2}(1+\sqrt{k})$ is the one supplied by Mohar's general estimate [12]

$$M_k(n) \le \frac{1+\sqrt{k}}{2}\, n, \tag{1.3}$$

which Mohar proved for graphs and for the wider class of symmetric matrices with entries in $[0,1]$, and which he asked to be sharpened. For $k = 3$ the reference coefficient is $(1+\sqrt{3})/2 = 1.3660254037\ldots$. The first explicit improvement below it is the estimate $\tau_3 \le 3/\sqrt{5} = 1.3416407864\ldots$ proved in Ebrahimi's thesis [5, Theorem 3.4.1]. The two-term matrix analogue, in which the entries are only required to lie in $[0,1]$, was treated by Ebrahimi, Mohar, Nikiforov and Ahmady [18].

The route that currently gives the strongest bounds was opened by Sivashankar, Tang and Wakhare [16]. They reduce the graph-eigenvalue problem to an extremal problem for orthogonal projections and then invoke the absolute projection constant $\gamma(r)$, normalised here by the requirement that

$$\|Q\|_1 \le \gamma(r)\, n \qquad \text{for every orthogonal projection } Q = Q^T = Q^2,\ \operatorname{tr} Q = r, \tag{1.4}$$

with $\|Q\|_1 = \sum_{ij} |q_{ij}|$ the entrywise ℓ_1 norm. Their two main estimates are

$$\lambda_k(G) \le \frac{\gamma(k-1)}{2(k-1)}\, n - 1, \qquad \Lambda_k(G) \le \frac{1+\gamma(k)}{2}\, n - k. \tag{1.5}$$

The second of these is their Proposition 6.1; together with the bound $\gamma(r) \le (r+\sqrt{r+2})/(1+\sqrt{r+2})$ of Deręgowska and Lewandowska [3], which implies $\gamma(r) < \sqrt{r}$ for every $r \ge 2$, it settles the conjecture (1.2). The first estimate in (1.5) yields the sharp individual bound $\lambda_3(G) \le n/3 - 1$; further recent work on λ_3 alone is due to Li [23] and to Leonida and Li [22], and lower bounds for the individual extrema c_k are surveyed in Linz [24]. For $k = 3$, using

the classical value $\gamma(3) = (1+\sqrt{5})/2$, the second estimate in (1.5) reads

$$\Lambda_3(G) \le \frac{3+\sqrt{5}}{4}\, n - 3 = 1.309016994374947\ldots\, n - 3. \tag{1.6}$$

For the two-term sum the situation is essentially complete: Kumar, Liu, Monterde, Pragada and Tait [8] proved $\Lambda_2(G) \le 8n/7$, Huang and Wei [7] determined the exact maximum together with the unique extremal graph, whose three part sizes are governed by n modulo 7, and Paulin [15] determined the closures of the rescaled pairs $\big((\lambda_1+1)/n, (\lambda_2+1)/n\big)$ and $\big((\lambda_2+1)/n, (\lambda_3+1)/n\big)$, from which $\lambda_1+\lambda_2 \le 8n/7-2$, $\lambda_3 \le n/3-1$ and $\lambda_2+\lambda_3 \le 2n/3-2$ all follow. None of these settles Λ_3, for the reason explained in the next section.

The graph-limit setting used throughout is standard; we follow Lovász [25] and Borgs, Chayes, Lovász, Sós and Szegedy [17].

1.3 Main results

For a finite simple undirected graph G of order $n \ge 3$, put

$$\Lambda_3(G) = \lambda_1(G) + \lambda_2(G) + \lambda_3(G),$$

where the adjacency eigenvalues are ordered algebraically. The signs in this sum are retained. The extremal problem asks how large $\Lambda_3(G)$ can be when only n is prescribed. The spectral-sum problem appears in Ebrahimi's thesis [5], whose Theorem 3.4.1 gives the explicit asymptotic coefficient $\limsup_n \max_{|G|=n} \Lambda_3(G)/n \le 3/\sqrt{5} < 1.3417$; general bounds for sums of leading eigenvalues were developed by Mohar [12]. More recently, Sivashankar, Tang and Wakhare [16, Proposition 6.1] proved

$$\Lambda_3(G) \le \frac{3+\sqrt{5}}{4} n - 3. \tag{1.7}$$

Their argument uses the rank-three absolute projection constant $\gamma(3) = (1+\sqrt{5})/2$; a proof of the requisite projection bound is provided by Deręgowska and Lewandowska [3].

The two quantities in that reduction are naturally coupled. If $X \in \mathbb{R}^3$ satisfies $\mathbb{E}XX^T = I_3$ and X' is an independent copy, they are

$$A = \mathbb{E}|X \cdot X'|, \qquad B = |\mathbb{E}X|^2.$$

The separate estimates $A \le (1+\sqrt{5})/2$ and $B \le 1$ give (1.7). A better graph bound therefore requires a quantitative restriction on their simultaneous size. The mixed projection constant expressing this question was already introduced in [16, Section 6.2]. We work directly with its isotropic-law formulation.

Define

$$c = \frac{106106902139519}{107310156800000} + \frac{41\sqrt{3}}{240} - \frac{53}{9000000} = 1.284673255723918505\ldots. \tag{1.8}$$

Theorem 1.1. *Every finite simple unweighted graph G on $n \geq 3$ vertices satisfies*

$$\Lambda_3(G) \leq cn - 3. \tag{1.9}$$

Moreover, every Borel probability law on $\mathbb{R}^3$ satisfying $\mathbb{E}XX^T = I_3$ obeys

$$\mathbb{E}|X \cdot X'| + |\mathbb{E}X|^2 \leq 2c. \tag{1.10}$$

Only a finite second moment is assumed; zero vectors and unbounded radii are allowed.

The coefficient decrease from (1.7) is $0.024343738651028919\ldots$, with the constant term -3 unchanged. This comparison concerns the stated universal order-only bound. It does not compare all estimates involving additional graph parameters, nor assert that c is optimal. For completeness we record the full chain of explicit coefficients: $3/\sqrt{5} = 1.341640786\ldots$ from [5, Theorem 3.4.1], then $(3+\sqrt{5})/4$ from [16, Proposition 6.1], and finally c.

Two adjacent results frame the standing of (1.9). For the two-term sum $S_2 = \lambda_1 + \lambda_2$, Huang and Wei [7] determine the exact maximum together with its unique extremal graph, whose three part sizes are governed by n modulo 7; Paulin [15] determines the closures of the rescaled pairs $\big((\lambda_1+1)/n, (\lambda_2+1)/n\big)$ and $\big((\lambda_2+1)/n, (\lambda_3+1)/n\big)$, giving $\lambda_1+\lambda_2 \leq 8n/7-2$, $\lambda_3 \leq n/3-1$ and $\lambda_2+\lambda_3 \leq 2n/3-2$. Neither settles the three-term sum, because the reduction behind (1.7) couples two quantities, and it is their joint control that forces the different geometry analysed here.

One prior announcement bears directly on our lower bound, and we record it here rather than in a footnote. The self-archived Zenodo deposits [13, 14] of Newton, dated 4 and 24 August 2026 and therefore earlier than this work, introduce the same extremal constant c_{w} under the name of a pole–pentagon law, together with the same optimising weight, and assert the matching upper bound $\Lambda_3(G) \leq c_{\mathrm{w}} n - 3$; the assertion is conditional by its own terms, since the final positivity layer is a machine certificate whose admissibility the author explicitly leaves to independent audit (Remark 3.3). We claim no priority for the constant. Theorem 1.1 is unconditional and self-contained, and is weaker in the coefficient by $c - c_{\mathrm{w}} = 0.003105058851750628\ldots$.

We use graphons to pass to an attained maximum. For a symmetric measurable kernel $W : [0,1]^2 \to [0,1]$, let

$$\Sigma_k(W) = \sup_{\mathbb{E}FF^T = I_k} \iint W(x,y) F(x) \cdot F(y)\, dx\, dy, \qquad \mathcal{M} = \sup_W \Sigma_3(W). \tag{1.11}$$

The supremum is over real orthonormal k-frames. Since $W \in L^2([0,1]^2)$, the operator T_W is Hilbert–Schmidt and therefore compact; it is also self-adjoint, so its spectrum is real and accumulates only at zero, and Σ_k is the sum of the first k positive eigenvalues, padded by zero if there are fewer than k of them. This convention applies only to graphons, not to $\Lambda_3(G)$. The maximum $\mathcal{M}$ is attained. For a maximizing graphon, choose an orthonormal top-three eigenframe $F = (f_1, f_2, f_3)^T$, with $f_1 \geq 0$, and call $F_{\#}dx$ its canonical law.

Theorem 1.2. *Let W attain $\mathcal{M}$, let $\sum_i p_i \delta_{v_i}$ $(p_i > 0)$ be its canonical law, and let L be the diagonal matrix of the three leading eigenvalues of W. Then:*

(a) the support has at most 24 points;

(b) W agrees almost everywhere with the binary step kernel $\mathbf{1}_{\{v_i \cdot v_j > 0\}}$ on the corresponding positive-measure fibers;

(c) the support points satisfy

$$\sum_i p_i v_i v_i^T = I_3, \qquad v_i^T L v_i = \mathcal{M}, \qquad v_{i1} > 0,$$
$$v_i \cdot v_j \neq 0, \qquad B_{ij} = \mathbf{1}_{\{v_i \cdot v_j > 0\}}, \qquad \sum_j p_j B_{ij} v_j = L v_i;$$

(d) for every positive-dot clique J in the support, the matrices $\{v_i v_i^T : i \in J\}$ are linearly independent; in particular $|J| \leq 6$.

This assertion concerns every original maximizer, not merely the existence of a maximizer with small support.

1.4 Method and relation to earlier work

The proof has two parts. First, second variation on the actual probability space gives a negative semidefinite kernel. On each of four sign classes, this controls the rank of the positive-part Gram kernel and forces finite canonical support. Moment-preserving variations then imply vector relations against every binary row of the maximizing kernel. Those rows separate distinct support points. A minimal-dependency argument shows that no second-moment relation can survive on a positive clique.

The variation, cloning, and orthant-reduction mechanisms have close antecedents in Kumar, Mohar, Mojallal and Pragada [9]. Related graphon ellipse and clique-cloning arguments for the two-eigenvalue problem appear in Kumar, Liu, Monterde, Pragada and Tait [8]. General bounds for minimal dependencies of powers of linear forms are due to Li and Liu [10]. Our separating-observation lemma is stated and proved with its specific hypotheses in Section 4.4; we do not claim a new general bound for such dependencies.

Second, for one isotropic law we retain the mean deficit, radial variance, regression error, and weighted harmonic moments of degrees three and four. A signed quartic majorant couples them in one exact identity. Directional Hermite inequalities and a radial comparison lead to an explicit sum of nonnegative terms. Harmonic moment and polynomial-majorant methods are classical; see Glazyrin [6], Deręgowska and Lewandowska [3, 4], and Sivashankar, Tang and Wakhare [16]. The contribution here is the particular joint control of these errors for the mixed signed and absolute energy, rather than the use of polynomial majorants by itself. The support theorem supplies a small additional term through self-pairs of actual atoms. Diagonal removal and effective-cardinality estimates are standard weighted-energy devices; see [1].

Chapters 3 and 4 prove the structural result; Chapters 5 and 6 give the joint moment certificate and transfer it to finite graphs. Chapter 7 reports the verification experiments and the exact scope of the bound, and Chapter 8 collects the open problems.

1.5 Organization of the thesis

Chapter 2 collects the standard material used later: the Ky Fan maximum principle, graphons and the cut metric, equitable partitions and blow-ups, isotropic laws and absolute projection constants, and the two harmonic tensor kernels g_3 and g_4 on the sphere. Nothing in that chapter is new, and every statement is cited.

Chapter 3 sets up the variational quantity $\mathcal{M} = \sup_W \Sigma_3(W)$ over graphons, proves that it is attained, gives the two-sided estimate $c_{\mathrm{w}} \le \mathcal{M} \le (1+\sqrt{3})/2$, and derives the four structural facts that every maximiser satisfies: the ellipsoid identity $F^T L F = \mathcal{M}$, the row-insertion inequality, the threshold form of the kernel, and the simplicity relations $\lambda_1 > \lambda_2 > \lambda_3 > \lambda_4$.

Chapter 4 proves that the canonical law of every maximiser is finitely supported, with at most 24 atoms. The argument runs through a second-variation inequality on the actual support, propagation of moment relations, a dimension descent over sign regions, and a separating family of binary observations that excludes minimal second-moment dependencies on positive cliques.

Chapter 5 builds the moment machinery for a single isotropic law on $\mathbb{R}^3$: the harmonic decompositions, the regression estimate, the signed quartic majorant f, and the exact joint defect identity that ties the mean deficit, the radial variance, the regression error and the two harmonic moments together.

Chapter 6 converts that identity into a coefficient bound. Directional Hermite inequalities and a radial comparison produce an explicit sum of nonnegative terms; a second pass extracts a reserve from the quartic slack; the support theorem contributes a positive atomic term; and the final certificate gives $\mathcal{M} \le c$. The chapter closes with the transfer to all isotropic laws and to finite graphs, the proof of Theorem 1.1, and the weighted-graph form of the bound.

Chapter 7 is computational. It records an exact-arithmetic verification protocol, an exhaustive check over all graphs of orders $3 \le n \le 7$, an exact spectral formula for the equal blow-ups of $K_1 \vee C_5$, convergence tables for the optimally weighted blow-ups, and a discussion of what these experiments do and do not establish.

Chapter 8 states the conclusions and lists the open problems, including the residual gap $c - c_{\mathrm{w}}$, the classification of extremal templates, and the adjacent two-term problems. Appendix A records which constants in the argument are forced and which were chosen and then verified.

1.6 Notation and conventions

A complete index is given in the List of Notation after the table of contents; the few conventions that matter globally are these. All graphs are finite, simple and undirected, and n always denotes $|V(G)|$ with $n \ge 3$. Eigenvalues are ordered algebraically, not by absolute value, and

Λ_k retains signs. The letter A denotes an adjacency matrix and T_W the integral operator of a graphon W. For a real symmetric matrix Q, $\|Q\|_1 = \sum_{ij} |Q_{ij}|$ is the entrywise ℓ_1 norm and $\|Q\|_F^2 = \mathrm{tr}(Q^2)$ is the squared Frobenius norm; these should not be confused with the operator norms $\|\cdot\|_{2\to 2}$ and $\|\cdot\|_\square$. The symbol $\mathbb{E}$ denotes expectation, and X' is always an independent copy of X. Rational numbers written with large numerators and denominators are exact, not rounded; rounded values are always followed by an ellipsis.

Four labels are used to distinguish the epistemic status of a statement. *Proved* means a complete argument appears in this thesis. *Verified* means the statement is an explicit identity or inequality that was re-checked in exact arithmetic. *Numerical* means floating point evidence only. *Gap* means an assertion that is neither proved nor verified here, and every gap is repeated in Chapter 8.

Chapter 2

Preliminaries

This chapter collects the standard tools used later. Nothing here is new; each statement is either proved in a few lines or cited. The purpose is to fix notation once and to isolate the two bridges on which the whole argument depends: the Ky Fan maximum principle, which turns a spectral sum into a trace against a rank-r projection, and the projection–isotropic-law bridge of Lemma 2.3, which turns such a trace into two moments of a probability law on $\mathbb{R}^r$.

2.1 The adjacency spectrum and Ky Fan's principle

Let $A = A_G$ be the adjacency matrix of a graph G of order n. Then A is real symmetric with zero diagonal, $\operatorname{tr} A = 0$, $\operatorname{tr} A^2 = 2|E(G)|$, and $\lambda_1(G) \le n-1$ with equality only for K_n. The spectrum of K_n is $(n-1, -1^{(n-1)})$, so

$$\Lambda_k(K_n) = n - k \qquad (1 \le k \le n), \tag{2.1}$$

and in particular $\Lambda_3(K_n) = n-3$; this is the trivial lower reference point for the coefficient problem, since any bound of the form $\Lambda_3(G) \le cn - 3$ is compatible with K_n exactly when $c \ge 1$. The spectrum of the cycle C_5 is $\big(2, \frac{\sqrt{5}-1}{2}, \frac{\sqrt{5}-1}{2}, -\frac{1+\sqrt{5}}{2}, -\frac{1+\sqrt{5}}{2}\big)$, and the join $K_1 \vee C_5$ has spectrum

$$1+\sqrt{6}, \quad \tfrac{\sqrt{5}-1}{2}, \quad \tfrac{\sqrt{5}-1}{2}, \quad 1-\sqrt{6}, \quad -\tfrac{1+\sqrt{5}}{2}, \quad -\tfrac{1+\sqrt{5}}{2},$$

so that

$$\Lambda_3(K_1 \vee C_5) = \sqrt{5} + \sqrt{6} = 4.6855577204\ldots. \tag{2.2}$$

Both facts follow from the equitable-partition calculation of Lemma 2.1 below with $q = 1$.

For a real symmetric matrix $M \in \mathbb{R}^{n\times n}$ let $\mathcal{P}_r(n) = \{Q \in \mathbb{R}^{n\times n} : Q = Q^T = Q^2,\ \operatorname{tr} Q = r\}$ be the set of orthogonal projections of rank r. Ky Fan's maximum principle [19] states that

$$\sum_{i=1}^{r} \lambda_i(M) = \max_{Q \in \mathcal{P}_r(n)} \operatorname{tr}(MQ), \tag{2.3}$$

where the maximum is attained, and $\lambda_1(M) \ge \cdots \ge \lambda_n(M)$. Every $Q \in \mathcal{P}_r(n)$ satisfies $Q_{ii} \ge 0$ and $\sum_i Q_{ii} = r$, and can be written $Q = YY^T$ with $Y \in \mathbb{R}^{n\times r}$ of orthonormal

columns. Throughout, $\|\cdot\|_{2\to2}$ denotes the operator norm and $\|\cdot\|_F^2 = \operatorname{tr}(M^2)$ the squared Frobenius norm.

2.2 Graphons, the cut metric and the quantities Σ_k

A *graphon* is a symmetric measurable function $W : [0,1]^2 \to [0,1]$. Since $W \in L^2([0,1]^2)$, the rule $(T_W f)(x) = \int_0^1 W(x,y)f(y)\,dy$ defines a self-adjoint Hilbert–Schmidt operator T_W on $L^2[0,1]$; hence T_W is compact, its spectrum is real, and it accumulates only at 0. List the nonzero eigenvalues as $\lambda_1(W) \geq \lambda_2(W) \geq \cdots \to 0$ together with the negative ones $\lambda_{-1}(W) \leq \lambda_{-2}(W) \leq \cdots \to 0$, and define

$$\Sigma_k(W) = \lambda_1(W) + \cdots + \lambda_k^+(W), \qquad \lambda_i^+(W) = \max(\lambda_i(W), 0), \tag{2.4}$$

that is, the sum of the first k positive eigenvalues, padded by zero if there are fewer than k of them. Equivalently, by (2.3),

$$\Sigma_k(W) = \sup_{\mathbb{E}FF^T = I_k} \iint W(x,y)\, F(x)\cdot F(y)\,dx\,dy,$$

the supremum ranging over measurable orthonormal k-frames $F = (f_1, \ldots, f_k)^T$. The quantity of interest is

$$\mathcal{M} = \sup_W \Sigma_3(W). \tag{2.5}$$

The padding convention applies to Σ_k only and never to $\Lambda_k(G)$.

For a graph G of order n the associated step graphon W_G takes the value A_{ij} on the square $[\frac{i-1}{n}, \frac{i}{n}) \times [\frac{j-1}{n}, \frac{j}{n})$. Identifying step functions constant on these intervals with vectors of $\mathbb{R}^n$ gives $T_{W_G} = A_G/n$, hence

$$\lambda_i(W_G) = \frac{\lambda_i(G)}{n}, \qquad \Sigma_k(W_G) = \frac{1}{n}\sum_{i=1}^{k}(\lambda_i(G))_+ \geq \frac{\Lambda_k(G)}{n}. \tag{2.6}$$

The inequality is strict precisely when $\lambda_k(G) < 0$. Consequently $\mathcal{M} \geq \limsup_n M_3(n)/n$, and the finite bound $\Lambda_3(G) \leq \mathcal{M}n$ is immediate; the difficulty, and the content of Chapter 6, is the exact constant term.

The *cut norm* of a bounded measurable $U : [0,1]^2 \to \mathbb{R}$ is $\|U\|_\square = \sup_{S,T}\left|\int_S\int_T U\right|$, the supremum over measurable $S, T \subset [0,1]$, and the *cut distance* between graphons is $\delta_\square(W_1, W_2) = \inf_\varphi \|W_1 - W_2^\varphi\|_\square$, the infimum over measure-preserving bijections φ of $[0,1]$. Two facts are used repeatedly and are standard [25, 17, 11]: the space of graphons is compact in $\delta_\square$; and for graphons W_1, W_2 with $D = W_1 - W_2$,

$$|\lambda| \leq 4\|D\|_\square\|f\|_\infty^2 \quad \text{and} \quad \|T_D\|_{2\to2}^3 \leq 4\|D\|_\square \tag{2.7}$$

for every eigenvalue λ of T_D with eigenfunction f. Since $|\Sigma_3(W_1) - \Sigma_3(W_2)| \leq 3\|T_D\|_{2\to2}$, the functional Σ_3 is continuous in $\delta_\square$; combined with compactness this gives attainment

of $\mathcal{M}$, and combined with the density of step graphons of graphs it shows that allowing general $[0,1]$-valued kernels creates no extremal graph that a genuine graph sequence cannot approach.

2.3 Equitable partitions, joins and blow-ups

Let G have a partition $V(G) = A_1 \sqcup \cdots \sqcup A_s$ such that every vertex of A_i has the same number q_{ij} of neighbours in A_j. The matrix $Q = (q_{ij})$ is the *quotient matrix* of the partition, the partition is *equitable*, and every eigenvalue of Q is an eigenvalue of A_G; the remaining eigenvalues of A_G come from the subspace of functions with zero sum on each part. A join $G_1 \vee G_2$ is the graph obtained by adding all edges between G_1 and G_2.

The following explicit case is used in Chapter 7.

Lemma 2.1 (Spectrum of an equal blow-up)**.** *Let H be a graph on s vertices and $q \geq 1$. Let $G = H[q]$ be obtained by replacing each vertex of H by a clique of size q and each edge of H by a complete bipartite graph. Then*

$$\operatorname{spec} G = \{\, q\lambda + q - 1 : \lambda \in \operatorname{spec} H \,\} \cup \{-1 \text{ with multiplicity } s(q-1)\}.$$

Proof. Index the vertices by pairs (i,a), $i \in [s]$, $a \in [q]$. Then

$$A_G = H \otimes J_q + I_s \otimes (J_q - I_q),$$

where J_q is the $q \times q$ all-ones matrix. Decompose $\mathbb{R}^{sq} = (\mathbb{R}^s \otimes \mathbf{1}_q) \oplus (\mathbb{R}^s \otimes \mathbf{1}_q^{\perp})$. Both summands are invariant. On the first, J_q acts as q and $J_q - I_q$ as $q-1$, so A_G acts as $qH + (q-1)I_s$. On the second, J_q acts as 0 and $J_q - I_q$ as $-I_q$, so A_G acts as $-I$. The two sums of multiplicities are s and $s(q-1)$, totalling sq. □

Remark 2.2. For unequal part sizes $q_1, \ldots, q_s$ the same decomposition applies with $D = \operatorname{diag}(q_1, \ldots, q_s)$: the mean-zero-within-parts subspace contributes -1 with multiplicity $\sum_i (q_i - 1)$, and the constant-on-parts subspace carries the symmetric quotient matrix

$$D^{1/2}(H + I_s)D^{1/2} - I_s. \tag{2.8}$$

This is the form used for the weighted blow-ups of $K_1 \vee C_5$ in Chapter 7.

2.4 Isotropic laws and the projection bridge

A Borel probability law μ on $\mathbb{R}^r$ is *isotropic* if $\int xx^T \, d\mu(x) = I_r$; write $X \sim \mu$ and let X' be an independent copy. Isotropy forces $\mathbb{E}|X|^2 = r$, $\mathbb{E}R^2 = r$ for $R = |X|$, and $|\mathbb{E}X|^2 \leq 1$, since $(v \cdot \mathbb{E}X)^2 \leq \mathbb{E}(v \cdot X)^2 = 1$ for every unit vector v. Zero vectors, atoms and unbounded radii are permitted.

The absolute projection constant $\gamma(r)$ is defined by (1.4). The classical values used here are $\gamma(2) = 4/3$ [3] and $\gamma(3) = (1 + \sqrt{5})/2$ [16, 3]; the general bound of Deręgowska and

Lewandowska [3] is

$$\gamma(r) \le \frac{r+\sqrt{r+2}}{1+\sqrt{r+2}} < \sqrt{r} \qquad (r \ge 2). \tag{2.9}$$

The strict inequality follows by writing $s = \sqrt{r+2}$, which gives $\gamma(r) \le s - 2/(1+s)$, and comparing $2/(s+\sqrt{r})$ with $2/(1+s)$.

Lemma 2.3 (Projection–isotropic-law bridge)**.** *Let $r \le n$ and let $A \in \mathbb{R}^{n\times n}$ be symmetric with $A_{ii} = 0$ and $0 \le A_{ij} \le 1$ for $i \ne j$. Let*

$$\Gamma_r = \sup_{\mu} \frac{\mathbb{E}|X \cdot X'| + |\mathbb{E}X|^2}{2}, \tag{2.10}$$

the supremum over isotropic laws on $\mathbb{R}^r$. Then

$$\sum_{i=1}^{r} \lambda_i(A) \le n\,\Gamma_r - r. \tag{2.11}$$

Moreover $\Gamma_r \le \frac{1}{2}(1+\gamma(r))$, so (2.11) recovers $\Lambda_r(A) \le \frac{1+\gamma(r)}{2}n - r$, and in particular $\Lambda_3(A) \le \frac{3+\sqrt{5}}{4}n - 3$ for $r = 3$.

Proof. By (2.3) and the hypotheses on A,

$$\sum_{i=1}^{r} \lambda_i(A) = \max_{Q\in\mathcal{P}_r(n)} \operatorname{tr}(AQ) \le \max_{Q\in\mathcal{P}_r(n)} \sum_{i\ne j} (Q_{ij})_+.$$

Since $Q_{ii} \ge 0$ and $\sum_i Q_{ii} = r$, and since $x_+ = (x+|x|)/2$,

$$\sum_{i\ne j} (Q_{ij})_+ = \frac{\|Q\|_1 + \mathbf{1}^T Q \mathbf{1}}{2} - r.$$

Write $Q = YY^T$ with orthonormal columns, let y_i be the rows of Y, and let X be uniformly distributed on $\sqrt{n}\,y_1, \dots, \sqrt{n}\,y_n$. Then $\mathbb{E}XX^T = \frac{1}{n}\sum_i n y_i y_i^T = I_r$, so X is isotropic, and

$$\mathbb{E}|X \cdot X'| = \frac{\|Q\|_1}{n}, \qquad |\mathbb{E}X|^2 = \frac{\mathbf{1}^T Q \mathbf{1}}{n}.$$

Substitution gives $\sum_{i\le r} \lambda_i(A) \le n\Gamma_r - r$. For the second assertion, $\mathbf{1}^T Q \mathbf{1} = \|Q\mathbf{1}\|_2^2 \le n$ and $\|Q\|_1 \le \gamma(r)n$ by (1.4), giving $\Gamma_r \le \frac{1}{2}(1+\gamma(r))$; for $r = 3$ use $\gamma(3) = (1+\sqrt{5})/2$. □

Remark 2.4. Lemma 2.3 uses nothing about A except the zero diagonal and the off-diagonal range $[0,1]$. It therefore applies to adjacency matrices of simple graphs, to $[0,1]$-weighted loopless graphs, and to any matrix in between; this is the mechanism behind [16, Proposition 6.1] and behind the matrix setting of [12]. Allowing weights above 1, or multiple edges, breaks it.

2.5 Two harmonic Gram kernels on the sphere

Let $S = S^2 \subset \mathbb{R}^3$. Define, for $u \in S$, the tensors

$$(\mathcal{L}_3(y))_{ijk} = \frac{y_i\delta_{jk} + y_j\delta_{ik} + y_k\delta_{ij}}{5}, \qquad \Psi_3(u) = u^{\otimes 3} - \mathcal{L}_3(u),$$

and, with $\Omega_{ijkl} = (\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})/15$ and, for symmetric traceless Z,

$$\begin{aligned}(\mathcal{L}_4(Z))_{ijkl} = \tfrac{1}{7}(&\delta_{ij}Z_{kl} + \delta_{ik}Z_{jl} + \delta_{il}Z_{jk}\\ &+ \delta_{jk}Z_{il} + \delta_{jl}Z_{ik} + \delta_{kl}Z_{ij}),\\ \Psi_4(u) = u^{\otimes 4} - \Omega - \mathcal{L}_4(uu^T - I_3/3),&\end{aligned}$$

the two scalar functions

$$g_3(t) = t^3 - \frac{3t}{5} = \frac{2}{5}P_3(t), \qquad g_4(t) = t^4 - \frac{6t^2}{7} + \frac{3}{35} = \frac{8}{35}P_4(t), \tag{2.12}$$

where P_3, P_4 are the Legendre polynomials. Direct expansion gives $\langle \Psi_3(u), \Psi_3(v)\rangle = g_3(u \cdot v)$ and $\langle \Psi_4(u), \Psi_4(v)\rangle = g_4(u \cdot v)$, with $\|\Psi_3(u)\|^2 = 2/5$ and $\|\Psi_4(u)\|^2 = 8/35$. Both g_3 and g_4 are therefore Gram kernels of explicit finite-dimensional maps and are positive semidefinite on $S \times S$; this is the concrete form of Schoenberg's criterion [29]. Two consequences are used later. For any $\mathbb{R}^3$-valued law,

$$\mathbb{E}[R\Psi_3(U)] \text{ and } \mathbb{E}[R\Psi_4(U)]$$

are well defined whenever $\mathbb{E}R^2 < \infty$, and both g_3 and g_4 admit exact polynomial majorants on $[-1, 1]$ that factor as $(1-t)(t-\frac{1}{2})^2$ times a positive factor; these are the identities (6.2) of Chapter 6. On the circle S^1 the analogous statement is that $t^4 - t^2 + \frac{1}{8}$ is positive semidefinite, because for $U = (\cos a, \sin a)$ it equals $\cos(4(a - a'))/8$.

2.6 Standing hypotheses

Unless a statement says otherwise: graphs are finite, simple, undirected and loopless, and $n = |V(G)| \geq 3$; graphons are symmetric measurable $[0, 1]$-valued kernels on $[0, 1]^2$; all probability laws are Borel on $\mathbb{R}^3$ and satisfy only $\mathbb{E}XX^T = I_3$ and hence $\mathbb{E}|X|^2 = 3$; eigenvalues are ordered algebraically; X' is an independent copy of X; and every displayed rational number is exact. Floating point values are reported with at least twelve significant digits and are always followed by an ellipsis when they are rounded.

Chapter 3

The graphon variational problem

The bound of Theorem 1.1 is proved by passing from graphs to graphons, solving an attained variational problem there, and transferring the answer back. This chapter carries out the first two steps. Section 1.6 and Chapter 2 have fixed the meaning of $\Sigma_3(W)$ and $\mathcal{M} = \sup_W \Sigma_3(W)$; here we establish that $\mathcal{M}$ is attained, locate it between two explicit constants, and extract from maximality every structural consequence that the later chapters consume.

The reason for working with graphons rather than with graphs of growing order is that Λ_3 is not continuous in any elementary graph metric, whereas Σ_3 is continuous in the cut metric by (2.7). Attainment then supplies an object to which calculus can be applied: a maximiser has a canonical eigenframe, and perturbing that frame inside the space of graphons produces exact first- and second-order identities. Those identities, and not any counting argument on finite graphs, are what make the structural results of this chapter and of Chapter 4 possible.

Two warnings about the logical order of the argument. First, no eigenvalue simplicity is assumed at the outset; the relations $\lambda_1 > \lambda_2$ and $\lambda_3 > \lambda_4$ are consequences, and they are derived in Lemma 3.4 and Lemma 3.6 by comparing genuinely different orthonormal frames. Second, the lower bound $c_{\mathrm{w}} \le \mathcal{M}$ is proved by an ordinary quotient computation for a join and is independent of the priority question discussed in Remark 3.3.

We may work on any standard nonatomic probability space. All changes of measure below can be realized on such a space.

Lemma 3.1. *Every $[0,1]$-valued graphon satisfies $\Sigma_2(W) \le 7/6$.*

Proof. We give the elementary argument for the rank-two absolute-energy bound needed below. Take any $\mathbb{R}^2$-valued law $\mathbb{E}YY^T = I_2$, write $Y = RU$ with a unit choice at zero, $C = \mathbb{E}R$, and $N = \mathbb{E}[RUU^T]$. Then $C^2 \le 2$ and $\operatorname{tr}(N^2) \le 1$, the latter from $\mathbb{E}\|U - NY\|^2 = 1 - \operatorname{tr}(N^2) \ge 0$. On the circle, $g_{4,2}(t) = t^4 - t^2 + 1/8$ is a positive kernel: if $U = (\cos a, \sin a)$, it is $\cos(4(a-a'))/8$. Hence its RR'-weighted integral is nonnegative. The exact majorant

$$\frac{2}{9} + \frac{11}{9}t^2 - \frac{4}{9}t^4 - |t| = \frac{(1-|t|)(2|t|-1)^2(|t|+2)}{9} \ge 0$$

therefore gives

$$\mathbb{E}|Y \cdot Y'| \le \frac{5}{18}C^2 + \frac{7}{9}\operatorname{tr}(N^2) \le \frac{4}{3}.$$

Also $|\mathbb{E}Y|^2 \le 1$. For any orthonormal two-frame in a graphon, $W(Y\cdot Y') \le (Y\cdot Y')_+$ pointwise, so its trace is at most $(4/3+1)/2 = 7/6$. Taking the supremum proves the lemma. The absolute-energy estimate is classical; see [6, Theorems 2 and 4]. □

Lemma 3.2. *The supremum $\mathcal{M}$ is attained, and*

$$c_{\mathrm{w}} := \frac{9+\sqrt{5}+2\sqrt{35}}{18} \le \mathcal{M} \le \frac{1+\sqrt{3}}{2}.$$

Every maximizing graphon has three positive leading eigenvalues.

Proof. For an orthonormal k-frame, $K(x,y) = F(x)\cdot F(y)$ satisfies $\int K = |\int F|^2 \le 1$ and $\int K^2 = k$. Hence $\int WK \le \int K_+ \le (1+\sqrt{k})/2$. For the lower bound, use a six-part kernel with template $K_1 \vee C_5$ and diagonal blocks equal to one. Give the center mass $a = 1-5b$ and each rim part mass b, where $0 < b < 1/5$. Put $\phi = (1+\sqrt{5})/2$. Its quotient spectrum is

$$\frac{1-2b \pm \sqrt{1+4b-36b^2}}{2}, \quad \phi b,\ \phi b,\ -b/\phi,\ -b/\phi.$$

Indeed the four nonconstant Fourier modes on the pentagon give the last four values; the remaining block is $\left(\begin{smallmatrix} a & \sqrt{5ab} \\ \sqrt{5ab} & 3b \end{smallmatrix}\right)$. Its positive eigenvalue exceeds $3b$, and its determinant is $-2ab < 0$. Consequently the sum of its three positive eigenvalues is

$$\frac{1+2\sqrt{5}b+\sqrt{1+4b-36b^2}}{2}.$$

Substitution of $b = (7+5\sqrt{7})/126$ gives c_{w}. This is the usual quotient calculation for a join construction; see [2]. In particular $c_{\mathrm{w}} > 32/25 > 7/6$.

For completeness we record why Σ_3 of this step kernel is exactly the sum of the three positive quotient eigenvalues. If f is supported in one part A_i and $\int_{A_i} f = 0$, then for $x \in A_i$ one has $(T_W f)(x) = \int_{A_i} f = 0$, while for x in any other part A_j one has $(T_W f)(x) = W_{ji}\int_{A_i} f = 0$; hence $T_W f = 0$. The whole block-mean-zero subspace therefore lies in the zero eigenspace, and Σ_3 is governed by the quotient matrix alone.

For attainment, let $D = W_1 - W_2$. If $\lambda \ne 0$ is an eigenvalue of T_D with unit eigenfunction f, then $|D| \le 1$ gives $\|f\|_\infty \le 1/|\lambda|$. The cut-norm dual estimate yields

$$|\lambda| = |\langle f, T_D f\rangle| \le 4\|D\|_\square \|f\|_\infty^2, \qquad \|T_D\|_{2\to 2}^3 \le 4\|D\|_\square.$$

The assertion is trivial when $T_D = 0$; otherwise compactness supplies an eigenvalue of modulus $\|T_D\|$. By the variational definition, $|\Sigma_3(W_1) - \Sigma_3(W_2)| \le 3\|T_D\|$. This also holds after relabeling. Cut-metric compactness [11] therefore gives a maximizer. Finally Lemma 3.1 implies $\lambda_3 \ge c_{\mathrm{w}} - 7/6 > 0$ at each maximizer. □

Remark 3.3. Priority for the constant c_{w}. The template $K_1 \vee C_5$ of Lemma 3.2 is the "pole–pentagon" law of the self-archived Zenodo deposits [13, 14], which appeared about two and a half months before the first version of the present paper. Writing the apex mass as $p^* = (91-25\sqrt{7})/126$ and each of the five rim masses as $r^* = (7+5\sqrt{7})/126$, so that

$p^* + 5r^* = 1$, those deposits obtain the identical value

$$\alpha_3 = \frac{1}{2} + \frac{\sqrt{5}}{18} + \frac{\sqrt{35}}{9} = \frac{9 + \sqrt{5} + 2\sqrt{35}}{18} = c_{\mathrm{w}}$$

at the identical optimising weight, and they further assert $\Lambda_3(G) \le \alpha_3 n - 3$ for every $n \ge 3$. That assertion is conditional. Its theorem is titled a *candidate* universal S_3 theorem, it is stated to hold "within the exact analytic and proof-carrying certificate chain", and the text declares that acceptance "properly remains subject to independent mathematical audit of the admissibility of the archived dual payments, the global sector partition and the exact verifier implementation". No refereed version is available. By contrast, the computation in Lemma 3.2 is an ordinary quotient calculation for a join and uses none of that machinery, so the lower bound $c_{\mathrm{w}} \le \mathcal{M}$ stands independently; we cite [13, 14] for the constant and for the naming and claim no priority for either. Nothing else in this paper depends on the unverified upper bound. Should it later be confirmed unconditionally, then $\mathcal{M} = c_{\mathrm{w}}$ and the interval displayed in Chapter efsec-further collapses to a point.

Fix a maximizing W and a top-three orthonormal eigenframe F, taking $f_1 \ge 0$. Write $L = \operatorname{diag}(\lambda_1, \lambda_2, \lambda_3)$. Taking the absolute value of a top Rayleigh maximizer supplies the nonnegative choice of f_1. We do not initially assume any eigenvalue simplicity.

Lemma 3.4. *Almost everywhere, $F^T L F = \mathcal{M}$. Moreover $\lambda_3 > \lambda_4$, where the fourth positive eigenvalue is zero if absent. The canonical support $S = \operatorname{supp}(F_{\#}dx)$ is compact, lies on the ellipsoid $v^T L v = \mathcal{M}$, and avoids the origin.*

Proof. For a bounded real g of mean zero, change density to $1 + tg$ for both signs of sufficiently small t. Set $Q = \int g F F^T$. The unchanged frame has complete Gram matrix $I + tQ$ and numerator matrix $L + t(QL + LQ) + O(t^2)$. After a common orthonormalization its trace is

$$\operatorname{tr}\big[(I + tQ)^{-1}\{L + t(QL + LQ) + O(t^2)\}\big] = \mathcal{M} + t \operatorname{tr}(LQ) + O(t^2).$$

Maximality gives $\int g F^T L F = 0$. Since every bounded mean-zero g is allowed, $F^T L F$ equals its mean $\mathcal{M}$ almost everywhere. This applies to every active eigenframe. If $\lambda_3 = \lambda_4 = \lambda > 0$, compare the frames with third entries f_3, f_4 and $(f_3 + f_4)/\sqrt{2}$, keeping f_1, f_2 fixed. The three identities imply $f_3^2 = f_4^2$ and $f_3 f_4 = 0$ almost everywhere, a contradiction. Only finitely many null sets are combined. Since $L \succ 0$, the ellipsoid is compact and excludes zero. It contains the full support, not just almost every frame value. □

Lemma 3.5. *For every measurable $h : [0,1] \to [0,1]$, the vector $r = \int hF$ satisfies $r^T L^{-1} r \le \mathcal{M}$.*

Proof. Give the original space A mass $u = 1 - \epsilon$ and a new independent part B mass ϵ; the kernel is W on $A \times A$, equals the prescribed cross-row h (a function on A) on $A \times B$ and $B \times A$, and is zero on $B \times B$. Put $t = L^{-1} r$ and $T = |t|^2$. For the trial frame take f_i on A and the constant t_i on B; then the $A \times B$ and $B \times A$ contributions to the numerator are $\epsilon u\, r_i t_j$ and its transpose. Dividing the entire frame by $\sqrt{u}$ leaves every Rayleigh quotient

unchanged, and the resulting Gram and numerator matrices are

$$G = I + (\epsilon/u)tt^T, \qquad N = uL + \epsilon(rt^T + tr^T).$$

The rank-one inverse formula gives the exact trial trace

$$\operatorname{tr}(G^{-1}N) = u\mathcal{M} + \frac{\epsilon r^T L^{-1} r}{1 + \epsilon T/u}.$$

This cannot exceed $\mathcal{M}$. Divide by ϵ and let $\epsilon \downarrow 0$. □

Lemma 3.6. *For $\nu = F_{\#}dx$ and every $v \in S$,*

$$\nu\{w : v \cdot w = 0\} = 0, \qquad \int_{v\cdot w>0} w \, d\nu(w) = Lv, \qquad v_1 > 0. \tag{3.1}$$

The original kernel is $\mathbf{1}_{\{F(x)\cdot F(y)>0\}}$ almost everywhere. It is irreducible, and $\lambda_1 > \lambda_2$.

Proof. Let $Z = \{\int hF : 0 \le h \le 1\}$ and $E = \{r : r^T L^{-1} r \le \mathcal{M}\}$. The set Z is compact by weak-star compactness and is contained in E by Lemma 3.5. Its support function is $H(v) = \int (v \cdot F)_+$. For almost every x, the original row has integral vector $LF(x)$, and $F(x)^T LF(x) = \mathcal{M}$. Since the support function of E is $\sqrt{\mathcal{M} v^T Lv}$, it follows that $H(F(x)) = \mathcal{M}$. Continuity then gives $H(v) = \mathcal{M}$ for every $v \in S$.

For such v, the unique support point of the strictly convex ellipsoid E in direction v is Lv. The row $h_v = \mathbf{1}_{\{v\cdot F>0\}}$ attains this support value, so $\int h_v F = Lv$. If $A_0 = \{v \cdot F = 0\}$, then $h_v + \mathbf{1}_A$ has the same support value for every measurable $A \subset A_0$. Uniqueness forces $\int_A F = 0$ for every such A, hence $F = 0$ almost everywhere on A_0. The ellipsoid identity implies that A_0 is null.

It remains to identify every maximizing row, not merely those obtained by perturbing h_v on A_0. Let $h \in [0, 1]$ satisfy $\int hF = Lv$ for some $v \in S$. Pairing with v and using $v^T Lv = \mathcal{M}$ gives $v \cdot \int hF = \mathcal{M}$, while on the other hand

$$v \cdot \int hF = \int h \, (v \cdot F) \, dx \le \int (v \cdot F)_+ \, dx = H(v) = \mathcal{M}.$$

Hence equality holds throughout. Since $0 \le h \le 1$, the pointwise bound $h \, (v \cdot F) \le (v \cdot F)_+$ is therefore an equality almost everywhere, which forces $h = 1$ almost everywhere on $\{v \cdot F > 0\}$ and $h = 0$ almost everywhere on $\{v \cdot F < 0\}$. As A_0 is null, $h = h_v$ almost everywhere. Thus the maximizing row is unique modulo null sets, which identifies every original row and proves the threshold formula.

If a nontrivial measurable cut had zero cross-kernel, its indicator multiplier would commute with T_W and with the isolated top-three spectral projection. The kernel $F(x) \cdot F(y)$ of that projection would vanish across the cut, contradicting the zero-equator assertion and Fubini. Thus W is irreducible. The zero set of the nonnegative f_1 would also give a zero cut by its eigen-equation, so $f_1 > 0$ almost everywhere. If another top eigenfunction orthogonal to f_1 existed, it would change sign. Equality in the absolute-value Rayleigh comparison would force zero edges between its positive and negative sets; its absolute value is a nonnegative

top eigenfunction and is positive almost everywhere. This contradicts irreducibility, proving simplicity. Finally, for every $v \in S$, $\lambda_1 v_1 = \int h_v f_1 > 0$: otherwise $h_v = 0$ almost everywhere and hence $Lv = 0$, which is impossible. □

Chapter 4

Finite canonical support

Lemma 3.6 identifies a maximising graphon almost everywhere with the binary threshold kernel $\mathbf{1}_{\{F(x)\cdot F(y)>0\}}$ built from its own canonical frame. The canonical law $\nu = F_{\#}dx$ is a priori an arbitrary Borel probability measure on the ellipsoid $\{v : v^T Lv = \mathcal{M}\}$, so the kernel could still be supported on a continuum of frame values. This chapter proves that it is not: every maximiser has a canonical law with at most 24 atoms, all of positive mass. That is Theorem 1.2, whose proof is assembled here from four steps.

The first step, Theorem 4.1, is a second-variation inequality. Multiplying the underlying measure by $1 + tg(F(x))$ with $\int g\, d\nu = 0$ is a legitimate change of graphon for small $|t|$, and maximality forces the second-order term to be nonpositive. The resulting kernel $K(v, w) = (v\cdot w)_+ - (v\cdot w)(v^T Lw)$ is negative semidefinite *on the actual support*; no extension to the whole sphere is claimed or needed.

The second step, Lemma 4.2, converts null vectors of that kernel into vector identities. On a set of mutually positively correlated points, a relation $\sum_i a_i v_i v_i^T = 0$ forces both $\sum_i a_i v_i = 0$ and $\sum_i a_i A(v_i, w) = 0$ for every w. Since the space of symmetric 3×3 matrices has dimension six and the sphere splits into four sign regions on which all dot products are positive, this bounds the dimension of the row space of A by 24 and provides actual anchor points $z_j \in S$.

The third step, Lemma 4.3 and Theorem 4.4, turns the dimension bound into a cardinality bound. Inside a single sign region of the anchor equators the rows $A(v, \cdot)$ are linear functions of v, so Lemma 4.3 applies and gives at most $2^{d-1} \le 4$ points per region; the number of regions of r lines is at most $1 + r(r + 1)/2$. This yields the intermediate bound $|S| \le 1204$.

The fourth step, Section 4.4, improves 1204 to 24 by excluding minimal second-moment dependencies on positive cliques. Lemma 4.6 is an elementary circuit argument that uses a separating family of binary observations; the actual rows B_{ki} of the maximiser supply such a family. The conclusion, Theorem 4.7, is that on every positive clique the map $a \mapsto \sum_{i\in J} a_i v_i v_i^T$ is injective, so a clique has at most six types; four sign regions then give $|S| \le 4 \cdot 6 = 24$.

4.1 A negative semidefinite kernel on the actual support

Continue with the maximizing graphon and canonical law of the previous section, and write $h_v(w) = \mathbf{1}_{\{v\cdot w>0\}}$. Define continuous symmetric kernels on $S \times S$ by

$$A(v,w) = (v\cdot w)_+, \qquad D(v,w) = (v\cdot w)(v^T L w), \qquad K = A - D.$$

Theorem 4.1 (Second variation on the actual support)**.** *For every finite list $v_1,\dots,v_s \in S$ and every real list $a_1,\dots,a_s$,*

$$\sum_{i,j} a_i a_j K(v_i,v_j) \le 0. \tag{4.1}$$

The assertion concerns only the actual canonical support.

Proof. For any bounded measurable real g on S with $\int g\,d\nu = 0$, use the legitimate density $1 + tg(F(x))$ for both signs of sufficiently small t. Set

$$Q = \int g(v) v v^T \, d\nu(v), \qquad R = \iint h_v(w) v w^T g(v) g(w)\, d\nu(v) d\nu(w).$$

The unchanged three coordinate functions have the complete Gram $G_t = I + tQ$ and numerator matrix $N_t = L + t(QL + LQ) + t^2 R$ in this changed measure. For small t, G_t is positive definite; multiplying the frame by $G_t^{-1/2}$ is a genuine common orthonormalization. Since $N_t = G_t L G_t + t^2(R - QLQ)$ and $\operatorname{tr}(LQ) = \mathcal{M}\int g = 0$, its exact trace is

$$\operatorname{tr}(G_t^{-1} N_t) = \mathcal{M} + t^2 \operatorname{tr}\big(G_t^{-1}(R - QLQ)\big) \le \mathcal{M}. \tag{4.2}$$

Letting $t \to 0$ gives $\iint K(v,w) g(v) g(w)\, d\nu(v) d\nu(w) \le 0$. There is no discarded remainder or varying-scheme constant in this calculation. By Lemma 3.4 and (3.1), $\int K(v,w) d\nu(w) = \mathcal{M} - \mathcal{M} = 0$ for every $v \in S$. Subtracting the mean therefore proves the same inequality for every bounded g. To obtain (4.1), combine repeated points and take disjoint shrinking neighborhoods of the remaining points. Each has positive ν-mass by full support. Use $g = \sum_i a_i 1_{U_i}/\nu(U_i)$, separately for each neighborhood size. This auxiliary g is bounded but need not have mean zero. The legal probability density direction is $g - \int g\,d\nu$; centering leaves the K quadratic form unchanged. Its bound need not be uniform across neighborhood sizes. Continuity of K on compact $S \times S$ gives (4.1) in the limit. This limiting step needs no uniform permissible size of t. No eigenvalue differentiation is used, including when $\lambda_2 = \lambda_3$. □

4.2 Propagation of moment relations

Lemma 4.2. *If $v_i \in S$ have $v_i \cdot v_j > 0$ for all i, j, then*

$$\sum_i a_i v_i v_i^T = 0 \quad \Longrightarrow \quad \sum_i a_i v_i = 0 \quad \textit{and} \quad \sum_i a_i A(v_i, w) = 0 \quad \textit{for every } w \in S.$$

Consequently the vector space spanned by the functions $A(v,\cdot)$, $v \in S$, has dimension $r \le 24$ and admits a basis $A(z_j,\cdot)$ with actual $z_j \in S$.

Proof. Write $Q_a = \sum a_i v_i v_i^T$ and $b_a = \sum a_i v_i$. The quadratic form in (4.1) is $|b_a|^2 - \operatorname{tr}(LQ_a^2)$ on this positive-dot set. For $Q_a = 0$ it is nonnegative, hence zero and $b_a = 0$. A null vector of a negative semidefinite quadratic form has zero cross terms: apply (4.1) also to this list with w appended and coefficient $s \in \mathbb{R}$. The resulting polynomial $2s\sum_i a_i K(v_i, w) + s^2 K(w, w)$ is nonpositive for all s, so its linear coefficient vanishes. The D sum vanishes because $Q_a = 0$; therefore the A sum vanishes for every w. Partition S by the signs of coordinates 2 and 3, assigning zero to the positive sign. Within each of the four disjoint sets, every dot product is strictly positive, since coordinate 1 is positive and the other two coordinate products are nonnegative. In each nonempty set choose a basis, of size at most 6, of its matrices vv^T in the six-dimensional space of symmetric 3 by 3 matrices. The implication just proved expresses every corresponding A row in the span of those basis-point rows. Combine the four lists and discard dependent rows. This proves $r \le 24$. These are spanning witnesses, not a replacement vertex measure. This bounds the row space, not yet the cardinality of the support. □

4.3 Sign regions and finite support

Lemma 4.3 (Sign-region dimension descent)**.** *Let $T \subset S$ have linear span of dimension $d \ge 1$. Suppose for every $w \in S$ there is a vector $b_w \in \mathbb{R}^3$ such that $A(v, w) = v \cdot b_w$ for all $v \in T$. Then $|T| \le 2^{d-1}$.*

Proof. If $d = 1$, the positive first coordinate and $v^T L v = \mathcal{M}$ make T a singleton. For general d, if no $w \in S$ has strictly positive dot product with one member of T and strictly negative dot product with another, any two members have the same threshold row outside the union of their two zero equators. Equation (3.1) then makes the members equal. Otherwise choose such a w, with witnesses v_+, v_-, and split $T_+ = \{v \in T : v \cdot w \ge 0\}$ and $T_- = \{v \in T : v \cdot w < 0\}$. On T_+, $v \cdot (b_w - w) = 0$; on T_-, $v \cdot b_w = 0$. Both are proper hyperplanes of $\operatorname{span} T$: evaluate $b_w - w$ at v_- and b_w at v_+. Thus both children have span dimension at most $d - 1$. They inherit the linearization for every $w' \in S$. Induction gives at most 2^{d-2} points in each child, proving the bound. Zeros were included in T_+, so none are lost in the dimension descent. □

Theorem 4.4 (Finite support of every maximizer)**.** *For every global maximizer of Σ_3, its actual canonical support satisfies*

$$|S| \le 4\big(1 + r(r+1)/2\big) \le 1204 =: N_*. \tag{4.3}$$

The original graphon agrees almost everywhere with a binary step graphon on these actual fibers, with positive masses, diagonal blocks 1, and no equal rows. For this particular maximizer $\min_{v,w \in S} |v \cdot w| > 0$.

Proof. Take the actual anchors $z_1, \dots, z_r$ from Lemma 4.2. Symmetry gives, for each $w \in S$, coefficients $c_j(w)$ with $A(v, w) = \sum_j c_j(w) A(v, z_j)$ for every $v \in S$. The union of the finitely many anchor equators has ν-measure zero by (3.1). On any fixed strict sign region of these anchors, $A(v, z_j)$ is either $v \cdot z_j$ or zero. Thus $A(v, w)$ is linear in v there, for every w, with coefficients determined by that region. Because $v_1 > 0$, use coordinates $(v_2/v_1, v_3/v_1)$. Each

anchor equator becomes an affine line, or is empty if its last two coordinates vanish. At most r lines have at most $1 + r(r+1)/2$ complementary regions: the j-th line has at most $j-1$ distinct previous intersection points and adds at most j regions. Coincident or parallel lines only lower this count. A strict sign pattern is an intersection of open half-planes, hence connected whenever nonempty. Apply Lemma 4.3 to $T = S$ intersected with each such sign region; its span dimension is at most 3, so it contains at most four points. We have therefore placed full ν-measure in a finite set of size bounded by (4.3). Its closure is the same finite set, so full support of ν implies that all of S is contained in it, including any previously excluded boundary candidates. Every support point is now an atom of positive mass. The individual zero-equator statements in (3.1) imply $v \cdot w \neq 0$ for every pair of support atoms. Taking a minimum over this finite set proves the asserted margin. The actual fibers $F^{-1}(v)$ provide the partition, and (3.1) identifies its rows. □

The margin asserted here may depend on the maximizer.

4.4 Separating binary observations

By the preceding section, write $S = \{v_1, \dots, v_m\}$ with actual masses $p_i > 0$. Put $V = (v_i^T)_{i=1}^m$, $P_0 = \operatorname{diag}(p_i)$ and $B_{ij} = \mathbf{1}_{\{v_i \cdot v_j > 0\}}$. Then

$$BP_0V = VL, \qquad \sum_i p_i v_i v_i^T = I_3, \qquad v_i^T L v_i = \mathcal{M}. \tag{4.4}$$

The full rows of B are distinct: equal rows would imply equal v_i by invertibility of L. A positive clique J means a subset of this actual support with $B_{ij} = 1$ for every $i, j \in J$. Extend vectors indexed by J by zero outside J.

Lemma 4.5 (Vector rigidity on a positive clique)**.** *For every real a supported on a positive clique, if $Q_a = \sum_i a_i v_i v_i^T = 0$, then*

$$\sum_i a_i = 0, \quad b_a := \sum_i a_i v_i = 0, \quad \sum_i B_{ki} a_i v_i = 0 \quad \textit{for every row } k. \tag{4.5}$$

Proof. Taking the trace against L gives $\mathcal{M} \sum a_i = 0$. For both signs of sufficiently small nonzero t, all $p_i + t a_i$ are positive. These are legitimate new fiber measures for the same binary template, realizable on an atomless space. They are not independent choices of moment data. The original coordinate functions have Gram matrix $I + tQ_a = I$ and compression

$$N(t) = L + t(Q_a L + L Q_a) + t^2 b_a b_a^T = L + t^2 b_a b_a^T.$$

The last term uses the clique assumption, and both cross terms use (4.4). Their trace cannot exceed the global maximum $\mathcal{M}$; hence $b_a = 0$ and $N(t) = L$. Equality of this trial trace with the maximum forces its three-dimensional space to be invariant. Explicitly, rotate one orthonormal coordinate into any unit perpendicular function; the first derivative of the trace gives zero off-space matrix element. This argument does not differentiate ordered eigenvalues.

The compression L then gives the eigenfunction equations at the perturbed measure. All fibers still have positive mass, so for every k, $\sum_i (p_i + ta_i)B_{ki}v_i = Lv_k$. Subtract (4.4) and divide by t to obtain (4.5). □

Moment-preserving cloning and the Ky Fan equality argument follow the mechanism in [9, Claims 2.2–2.5]. The permissible size of t may depend on the maximizer and the chosen direction.

4.4.1 A separating observation lemma

Minimal dependencies of rank-one powers have a broader theory; see Li and Liu [10]. The following lemma uses the additional binary separation hypothesis. Its elementary proof is included.

Lemma 4.6 (Moment independence from separating binary observations)**.** *Let $T = \{v_i : i \in J\}$ be a finite set of distinct points in $\mathbb{R}^3$ with $v_{i1} > 0$ and $v_i^T L v_i = \mathcal{M} > 0$, where L is positive definite. Let $\mathcal{B}$ be a family of functions $b : J \to \{0, 1\}$ containing the constant-one function and separating every pair of distinct indices. Suppose, for every real a with $\sum_{i\in J} a_i v_i v_i^T = 0$, that*

$$\sum_{i\in J} a_i b(i) v_i = 0 \quad \text{for every } b \in \mathcal{B}. \tag{4.6}$$

Then the matrices $\{v_i v_i^T : i \in J\}$ are linearly independent.

Proof. Suppose instead that $I \subseteq J$ is a support-minimal dependent family of these matrices, with relation $a_i \neq 0$ for all $i \in I$. Write $s = |I|$. Since $\dim \mathrm{Sym}_3 = 6$, minimality gives $s \leq 7$. The ellipse equation gives $\sum_I a_i = 0$, while the constant observation in (4.6) gives $\sum_I a_i v_i = 0$. With $w_i = (1, v_i)^T \in \mathbb{R}^4$, all three identities combine to give

$$\sum_{a_i>0} a_i w_i w_i^T = \sum_{a_i<0} (-a_i) w_i w_i^T.$$

Both sides are positive semidefinite. The range of a positive weighted sum of $w_i w_i^T$ is precisely the span of those w_i: its kernel consists of vectors perpendicular to every such w_i. Consequently the two spans coincide, and the affine hull H of all v_i in the circuit satisfies

$$1 + \dim H = \dim \operatorname{span}\{w_i : i \in I\} \leq \min(|\{a_i > 0\}|, |\{a_i < 0\}|) \leq 3.$$

This is an elementary positive-Gram range argument, not a new general circuit bound. Distinct points here cannot be proportional: equal ellipse radius forces the proportionality factor to be ± 1, and the positive first coordinate excludes minus one. Also an affine line meets this positive-definite ellipsoid in at most two points, by its strictly quadratic restriction to the line. If $\dim H \leq 1$, there are at most two points, whose rank-one matrices are independent by nonproportionality, a contradiction. It remains that $\dim H = 2$.

If $0 \in H$, this is a two-dimensional linear plane. Its rank-one matrices span a space of dimension at most three, so minimality now gives $s \leq 4$. For any observation b nonconstant

on I, (4.6) and the total vector relation give nonzero-coefficient linear relations on both nonempty sets $I_1 = \{i : b(i) = 1\}$ and $I_0 = \{i : b(i) = 0\}$. Nonproportionality forces each set to contain at least three points. This would give $s \geq 6$, a contradiction. If instead every observation in $\mathcal{B}$ were constant on I, then minimality excludes $s = 1$ (a one-element circuit is impossible because $v_i v_i^T \neq 0$), so we may choose distinct $i, j \in I$; since $\mathcal{B}$ separates every pair of distinct indices of J, some $b \in \mathcal{B}$ satisfies $b(i) \neq b(j)$, contradicting constancy on I.

If $0 \notin H$, write $H = \{v : c \cdot v = 1\}$. Any three distinct points of H on the ellipsoid are linearly independent: a linear relation has coefficient sum zero after applying c, hence would make the three points affinely collinear, which the line-intersection fact forbids. A nonconstant observation would therefore require at least four points on each of I_1, I_0, yielding $s \geq 8$, again a contradiction. Thus every observation is constant on I. A one-element circuit is impossible because $v_i v_i^T \neq 0$; separation of any two indices in I is contradicted. □

Theorem 4.7 (Every actual positive clique has at most six types)**.** *For every actual maximizer, and every positive clique J of its full support, the map $a \mapsto \sum_{i \in J} a_i v_i v_i^T$ is injective. In particular $|J| \leq 6$. If $\{v_i : i \in J\}$ lies in a linear plane, then $|J| \leq 3$. If it lies in an affine plane not containing zero, then $|J| \leq 5$.*

Proof. Use the actual observations $b_k(i) = B_{ki}$, for all rows k. For nonempty J, a row $k \in J$ supplies the constant-one observation. They separate indices: B is symmetric and has distinct full rows. Lemma 4.6 now applies by (4.5). The linear-plane refinement follows from $\dim \mathrm{Sym}_2 = 3$. For an affine plane $c \cdot v = 1$, the nonzero symmetric matrix $L - \mathcal{M} c c^T$ is orthogonal to every $v_i v_i^T$. It is nonzero because L has rank three while $\mathcal{M} c c^T$ has rank at most one. This leaves dimension at most five. Empty cliques require no argument. □

Corollary 4.8 (Proof of Theorem 1.2)**.** *Every maximizer has at most 24 distinct canonical support points.*

Proof. Partition its full support by the signs of coordinates two and three, assigning a zero coordinate to the positive sign. There are at most four parts. Within each part, the last two coordinate products are nonnegative and the first-coordinate product is strictly positive. Each part is a positive clique. Theorem 4.7 gives $m \leq 4 \cdot 6 = 24$ on its original support. This proves Theorem 1.2. □

Chapter 5

A joint moment identity for one isotropic law

Chapters 3 and 4 reduced the structural problem to a single probability law. This chapter and the next bound it. From here to the end of Proposition 6.7 the object of study is an arbitrary Borel probability law on $\mathbb{R}^3$ with $\mathbb{E}XX^T = I_3$; the maximising graphon enters only at the very end, in Section 6.3, where its canonical law is shown to satisfy the extra radial hypothesis that the argument needs.

The two quantities to be controlled are

$$A = \mathbb{E}|X \cdot X'|, \qquad B = |\mathbb{E}X|^2,$$

and the target is $A + B \le 2c$. The classical route bounds them separately: $A \le \gamma(3) = (1+\sqrt{5})/2$ and $B \le 1$, giving $A + B \le (3+\sqrt{5})/2$ and therefore the coefficient $(3+\sqrt{5})/4$. That route is exhausted, because both individual bounds are sharp and are attained by different laws. Any improvement must exploit the fact that the two quantities cannot be large simultaneously.

The device used here is a single exact identity rather than a chain of estimates. Section 5.1 fixes the notation and decomposes the law into a radial part $R = |X|$, a directional part $U = X/R$, the mean $\mathbf{m} = \mathbb{E}X$, the second-moment tensor $M = \mathbb{E}[RUU^T] = \frac{C}{3} I_3 + Z$, and the regression vector $\xi = U - MX$. It records the scalar budget $D = e + \alpha V + \beta J + \rho K^2 + S$ coming from the classical absolute-value majorant h_0, and proves a regression estimate that bounds the weighted linear moment $L = \mathbb{E}[R(t - q_0)^2]$ in terms of e, V, J with an explicit nonnegative remainder r_{reg}.

Section 5.2 replaces h_0 by a signed quartic. The polynomial f is the degree-four Hermite interpolant with double contact to $|w| + w$ at $w = -3/8$ and $w = 1/2$ and a contact at $w = 1$; its slack $\sigma = f - |w| - w$ is nonnegative on $[-1, 1]$ and, crucially, dominates the explicit quadratic $(1 - 9w^2)/25$. Integrating the expansion of f against the harmonic decompositions produces the single exact identity (5.7),

$$D = \delta_{\mathrm{q}} + b_e e + b_V V + b_J J + a_3 H^2 + a_4 K^2 + P,$$

in which the mean deficit e, the radial variance V, the regression error J, the two harmonic moments H, K and the slack moment P all appear at once. Lemma 5.2 then extracts from P the linear payment $\frac{3}{50}(V - 9J)$ plus a nonnegative remainder r_{rad}, under a bounded-radius hypothesis with endpoint ratio at most 3.

Nothing in this chapter is optimised yet; the identity is exact and every remainder is retained. The next chapter pays for the coefficient.

5.1 Moments and regression

We now turn to the numerical estimate. Throughout this section the law is arbitrary and isotropic. Bounded positive radii will be imposed only when a radial comparison requires them. For later bookkeeping define

$$d_0 = -\frac{25624284539519}{53655078400000} + \frac{\sqrt{5}}{2} - \frac{41\sqrt{3}}{120}, \qquad d_1 = d_0 + \frac{103}{9000000}, \qquad d = d_1 + \frac{1}{3000000}. \tag{5.1}$$

Then $d = (3 + \sqrt{5})/2 - 2c$, with c as in (1.8). The intermediate constants separate the joint moment estimate from the final atomic term. Fix $\mathbb{E}XX^T = I_3$, let $R = |X|$, and put $U = X/R$ when $R > 0$. At $R = 0$ choose any fixed unit vector for U. Define

$$C = \mathbb{E}R, \quad V = 3 - C^2, \quad \mathbf{m} = \mathbb{E}X, \quad B = |\mathbf{m}|^2, \quad e = 1 - B,$$
$$A = \mathbb{E}|X \cdot X'|, \qquad D = \frac{3 + \sqrt{5}}{2} - A - B.$$

Here $V, e \geq 0$: $\mathbb{E}R^2 = 3$ and Cauchy–Schwarz gives $(v \cdot \mathbf{m})^2 \leq \mathbb{E}(v \cdot X)^2 = 1$ for every unit v. All pair expectations below are over independent copies of this same law. Set

$$M = \mathbb{E}[RUU^T] = \frac{C}{3} I_3 + Z, \qquad J = 1 - \mathrm{tr}(M^2) = \frac{V}{3} - \|Z\|_F^2.$$

The regression vector $\xi = U - MX$ satisfies

$$\mathbb{E}[X\xi^T] = 0, \quad \mathbb{E}\|\xi\|^2 = J \geq 0, \quad \mathbb{E}UU^T = M^2 + \mathbb{E}[\xi\xi^T]. \tag{5.2}$$

In particular J is an actual regression error, not a freely chosen moment.

Here are the harmonic normalizations used throughout. All tensor inner products sum over all ordered indices. Put

$$\Omega_{ijkl} = \frac{\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}}{15}, \qquad (\mathcal{L}_3(y))_{ijk} = \frac{y_i\delta_{jk} + y_j\delta_{ik} + y_k\delta_{ij}}{5}.$$

For symmetric traceless Z, put

$$(\mathcal{L}_4(Z))_{ijkl} = \frac{1}{7}(\delta_{ij}Z_{kl} + \delta_{ik}Z_{jl} + \delta_{il}Z_{jk} + \delta_{jk}Z_{il} + \delta_{jl}Z_{ik} + \delta_{kl}Z_{ij}).$$

For unit u, set

$$\Psi_3(u) = u^{\otimes 3} - \mathcal{L}_3(u), \qquad \Psi_4(u) = u^{\otimes 4} - \Omega - \mathcal{L}_4(uu^T - I_3/3),$$
$$T_i = \mathbb{E}[R\Psi_i(U)], \qquad H = \|T_3\|_F, \qquad K = \|T_4\|_F.$$

Contraction in two indices gives the orthogonal decompositions

$$\mathbb{E}[RU^{\otimes 3}] = T_3 + \mathcal{L}_3(\mathbf{m}), \qquad \mathbb{E}[RU^{\otimes 4}] = T_4 + C\Omega + \mathcal{L}_4(Z).$$

Indeed $\operatorname{ct}\Omega = I_3/3$, $\|\Omega\|^2 = 1/5$, $\operatorname{ct}\mathcal{L}_4(Z) = Z$, and $\langle T, \mathcal{L}_4(Z)\rangle = (6/7)\langle \operatorname{ct} T, Z\rangle$. Also $\operatorname{ct}\mathcal{L}_3(y) = y$ and $\langle N, \mathcal{L}_3(y)\rangle = (3/5)\operatorname{ct} N \cdot y$. It follows directly that

$$\begin{aligned} &\langle \Psi_3(u), \Psi_3(v)\rangle = g_3(u\cdot v), && g_3(t) = t^3 - 3t/5, && \|\Psi_3(v)\|^2 = 2/5, \\ &\langle \Psi_4(u), \Psi_4(v)\rangle = g_4(u\cdot v), && g_4(t) = t^4 - 6t^2/7 + 3/35, && \|\Psi_4(v)\|^2 = 8/35. \end{aligned} \tag{5.3}$$

The tensor moments involve R, not R^3 or R^4, and are integrable under $\mathbb{E}R^2 = 3$.

Write $s = \sqrt{5}$ and define the scalar absolute-value majorant and its slack by

$$a = \frac{1+s}{6}, \quad b = \frac{7s-5}{8}, \quad \rho = \frac{15s-25}{16}, \qquad h_0(z) = \frac{(1-z)(sz-1)^2(sz+s+2)}{2(s+1)^2}.$$

Expansion gives, for $|t| \le 1$,

$$a + b(t^2 - 1/3) - \rho(t^4 - 1/5) - |t| = h_0(|t|) \ge 0.$$

This majorant is from [16, Lemma 4.4]. Let $S = \mathbb{E}[RR'h_0(|U\cdot U'|)]$. The decompositions above give

$$A = aC^2 + b\|Z\|^2 - \rho\left(K^2 + \frac{6}{7}\|Z\|^2\right) - S.$$

Consequently

$$D = e + \alpha V + \beta J + \rho K^2 + S, \qquad \alpha = \frac{2s-1}{14}, \quad \beta = \frac{s+10}{14}. \tag{5.4}$$

Useful rational bounds are $\alpha \ge 26/105$, $\beta \ge 6/7$, and $\rho \ge 1/2$, using $s > 67/30$, $s \ge 2$, and $s \ge 11/5$.

5.1.1 A regression estimate

Suppose $B > 0$. Put

$$x = \sqrt{B}, \quad v = \mathbf{m}/x, \quad t = U\cdot v, \quad Y = Rt,$$
$$q_0 = 1/\sqrt{3}, \quad q = C/(3x), \quad L = \mathbb{E}[R(t-q_0)^2].$$

Then $\mathbb{E}Y = x$, $\mathbb{E}Y^2 = 1$, $\operatorname{Var} Y = e$. If $w_0 = \mathbb{E}\xi$, (5.2) gives $w_0 = \mathbb{E}[(1 - \mathbf{m}\cdot X)\xi]$, hence $|w_0|^2 \le eJ$. For traceless symmetric 3 by 3 matrices, $|Zv|^2 \le (2/3)\|Z\|^2$; the other two eigenvalues accompanying an eigenvalue z have squared sum at least $z^2/2$. Writing

$J_v = \mathbb{E}(v \cdot \xi)^2 \le J$, we obtain

$$\begin{aligned}\mathbb{E}(t-q)^2 &= |Mv - q\mathbf{m}|^2 + q^2 e + J_v - 2qv \cdot w_0 \\ &\le 2V/9 + J/3 + q^2 e + 2q\sqrt{eJ} =: F_0, \\ \mathbb{E}(Y - qR)^2 &\le V/3 + 3q^2 e + 2q\sqrt{eV} =: G_0.\end{aligned}$$

Here $Mv - q\mathbf{m} = Zv$ and $\mathbb{E}RY \ge Cx - \sqrt{eV}$. Cauchy–Schwarz gives $L_q := \mathbb{E}[R(t-q)^2] \le \sqrt{F_0 G_0}$. Note that $Rt = R(U \cdot v) = (RU) \cdot v = X \cdot v = Y$, so $\mathbb{E}[Rt] = \mathbb{E}Y = x$; this is the only place at which the regression vector enters the exact identity below. With $\zeta = C/\sqrt{3}$,

$$L_q - L = q_0(\zeta - x)^2(\zeta + 2x)/x^2 \ge 0.$$

Thus, writing $u = \sqrt{e}$, $z = \sqrt{V}$, $j = \sqrt{J}$,

$$L \le \overline{L} := \frac{3}{4}F_0 + \frac{1}{3}G_0 = \frac{7q^2}{4}u^2 + \frac{5}{18}z^2 + \frac{1}{4}j^2 + \frac{2q}{3}uz + \frac{3q}{2}uj. \tag{5.5}$$

Set $r_{\mathrm{reg}} = \overline{L} - L \ge 0$. This remainder will be kept in the joint identity. No lower radius bound has been used here.

5.2 A signed quartic and a joint defect identity

Define

$$f(w) = \frac{8352 + 43806w + 54016w^2 - 12928w^3 - 10240w^4}{41503}, \qquad \sigma(w) = f(w) - |w| - w.$$

The polynomial is the degree-at-most-four Hermite interpolant with double contacts to $|w|+w$ at $-3/8$ and $1/2$, and a contact at 1. Uniqueness follows because a difference of two such polynomials would have five zeros counted with multiplicity. The following explicit identities prove the needed inequalities; interpolation alone is not a majorant proof.

Lemma 5.1. *For every $w \in [-1, 1]$,*

$$\sigma(w) \ge 0, \qquad \sigma(w) \ge \frac{1 - 9w^2}{25}. \tag{5.6}$$

Proof. Direct expansion gives

$$41503\sigma(w) = \begin{cases}(1-w)(2w-1)^2(8352 + 2560w), & 0 \le w \le 1, \\ (8w+3)^2(928 - 82w - 160w^2), & -1 \le w \le 0.\end{cases}$$

The final factor on the negative interval is concave and at least 850. This proves nonnegativity, including the join at 0 and both endpoints. The second assertion follows immediately when $|w| \ge 1/3$. For $w = -z$, $0 \le z \le 1/3$, write

$$\sigma(-z) = (z - 3/8)^2 \frac{59392 + 5248z - 10240z^2}{41503} \ge \frac{10}{7}(z - 3/8)^2.$$

Indeed $z(5248 - 10240z) \geq 0$ on this interval and $59392/41503 > 10/7$. The remaining comparison is the exact square

$$\frac{10}{7}(z - 3/8)^2 - \frac{1 - 9z^2}{25} = \frac{313}{175}\left(z - \frac{375}{1252}\right)^2 + \frac{109}{250400} > 0.$$

For $w = z$, $0 \leq z \leq 1/3$, the positive-interval factorization gives

$$\sigma(z) = (z - 1/2)^2 \frac{(1 - z)(33408 + 10240z)}{41503} \geq \frac{1}{2}(z - 1/2)^2,$$

because the coefficient is at least $22272/41503 > 1/2$. Now use

$$\frac{1}{2}(z - 1/2)^2 - \frac{1 - 9z^2}{25} = \frac{43}{50}\left(z - \frac{25}{86}\right)^2 + \frac{53}{4300} > 0.$$

These intervals exhaust $[-1, 1]$. □

Put $P = \mathbb{E}[RR'\sigma(U \cdot U')]$ for the same actual law as before, and set

$$a_3 = \frac{12928}{41503}, \quad a_4 = \frac{10240}{41503}, \quad b_e = \frac{16386}{18865},$$
$$b_V = \frac{64608}{290521}, \quad b_J = \frac{316672}{290521}, \quad \delta_{\mathrm{q}} = \frac{3 + \sqrt{5}}{2} - \frac{544886}{207515}.$$

Integration of the expanded f, using (5.3), gives the single exact joint budget

$$D = \delta_{\mathrm{q}} + b_e e + b_V V + b_J J + a_3 H^2 + a_4 K^2 + P. \tag{5.7}$$

For example, writing $f = c_0 + c_1 w + c_2 w^2 - a_3 w^3 - a_4 w^4$, the coefficients are $b_e = c_1 - 3a_3/5$, $b_V = c_0 + 3a_4/35$, $b_J = c_2 - 6a_4/7$, and the integrated constant is $3(c_0 - a_4/5) + b_e + c_2 = 544886/207515$. All tensors are R-weighted and σ is bounded, so this equality uses no radial moment beyond the second one. Identity (5.4) will only justify a small-D branch; it is not added as an independent resource to (5.7).

Lemma 5.2. *Suppose $0 < r_- \leq R \leq r_+ < \infty$ almost surely and $r_+/r_- \leq 3$. Then*

$$P = \frac{3}{50}(V - 9J) + r_{\mathrm{rad}}, \qquad r_{\mathrm{rad}} \geq 0. \tag{5.8}$$

Proof. For $w = U \cdot U'$, the exact pair identities from the full Gram are

$$V = \frac{1}{2}\mathbb{E}(R - R')^2, \qquad J = \frac{1}{2}\mathbb{E}[(R - R')^2 w^2].$$

The latter uses $\mathbb{E}[R^2 w^2] = \mathbb{E}[(R')^2 w^2] = 1$ and $\mathbb{E}[RR' w^2] = \mathrm{tr}(M^2)$; it assumes no independence of radius and direction. The ratio bound gives $(R - R')^2 \leq (4/3)RR'$. Use the two conclusions of Lemma 5.1 in this order:

$$P \geq \frac{3}{4}\mathbb{E}[(R - R')^2 \sigma(w)] \geq \frac{3}{100}\mathbb{E}[(R - R')^2(1 - 9w^2)] = \frac{3}{50}(V - 9J).$$

The expression $V - 9J$ need not be nonnegative. Define r_{rad} as the actual difference; only this difference is claimed nonnegative. $\square$

Chapter 6

The coefficient bound and its transfer

The exact identity (5.7) is now converted into a lower bound for the defect $D = (3+\sqrt{5})/2 - A - B$, which by Section 6.3 equals $(3+\sqrt{5})/2 - 2\mathcal{M}$ for the canonical law of a maximiser. Three ingredients are combined: directional Hermite inequalities that pay for the harmonic moments H^2 and K^2, a radial comparison that turns the slack moment P into a linear payment in V and J, and an atomic term supplied by the finite support theorem of Chapter 4. Each ingredient leaves a nonnegative remainder, and the point of the construction is that all of them are kept in one identity rather than discarded.

Section 6.1 carries out the directional step and produces the intermediate certificate (6.3) with coefficient d_0. Section 6.2 finds a reserve inside the quartic slack by means of the cubic-free contact polynomial z_*, improving d_0 to d_1. Section 6.3 verifies the radial hypothesis for an actual maximiser, adds the positive-mass self-pair term, and reaches the final constant d, equivalently $\mathcal{M} \le c$; it then transfers the bound to all isotropic laws and to finite graphs, proving Theorem 1.1, and records the exact scope of the result.

6.1 Directional and radial control

Throughout this section assume $0 < r_- \le R \le r_+ < \infty$ almost surely, with $r_+/r_- \le 3$, and work in the branch $D < 1/20$. Equation (5.4) implies

$$e < 1/20, \quad x > 9/10, \quad V < 21/104 < 2/9, \quad C > 5/3, \qquad q^2 = \frac{3-V}{9(1-e)} \le \frac{20}{57}. \tag{6.1}$$

In particular $\mathbf{m} \neq 0$. Put $p_i = \langle T_i, \Psi_i(v)\rangle = \mathbb{E}[Rg_i(t)]$. The elementary Hermite identities

$$\begin{aligned} \Big(\frac{53}{28}t^2 - \frac{9}{4}t + \frac{41}{70}\Big) - g_4(t) &= (1-t)(t-1/2)^2(t+2) \ge 0, \\ (2t^2 - 37t/20 + 1/4) - g_3(t) &= (1-t)(t-1/2)^2 \ge 0 \end{aligned} \tag{6.2}$$

hold on $[-1,1]$. Define

$$\gamma_4 = 9/4 - 53\sqrt{3}/42, \quad \ell_4 = 9/4 - 511\sqrt{3}/420, \quad \ell_3 = 37/20 - 11\sqrt{3}/12.$$

Here $0 < \gamma_4 < 1/15$ (square the positive rational comparisons). Since $\mathbb{E}Rt = x$, $\mathbb{E}Rt^2 = L + 2x/\sqrt{3} - C/3$, the first identity gives

$$-p_4 \geq \gamma_4 x + 19C/420 - 53L/28 \geq \ell_4 - e/15 - 19V/1400 - 53L/28.$$

We used $x \geq 1 - e$ and $C \geq \sqrt{3} - 3V/10$; the latter follows from $V = (\sqrt{3} - C)(\sqrt{3} + C)$ and $C > 5/3$. The second identity gives

$$-p_3 \geq (37/20 - 4/\sqrt{3})x + 5C/12 - 2L \geq \ell_3 + 9e/40 - V/8 - 2L.$$

Here $4/\sqrt{3} - 37/20 > 9/20$ and $x \leq 1 - e/2$. Define the nonnegative differences in these last two inequalities by

$$\begin{aligned} r_4 &= -p_4 - \ell_4 + e/15 + 19V/1400 + 53L/28, \\ r_3 &= -p_3 - \ell_3 - 9e/40 + V/8 + 2L. \end{aligned}$$

The polynomial identities, rather than a fixed sphere assumption, justify these directional estimates for the actual weighted moments.

Set $\eta = 1/10$, $\theta = 6/25$, and $\ell = 53\eta/28 + 2\theta = 937/1400$. Define the tensor squares

$$\mathcal{R}_4 = a_4 \left\| T_4 + \frac{\eta}{2a_4}\Psi_4(v) \right\|^2, \qquad \mathcal{R}_3 = a_3 \left\| T_3 + \frac{\theta}{2a_3}\Psi_3(v) \right\|^2.$$

Their exact expansions give

$$a_4 K^2 = -\eta p_4 - \frac{2\eta^2}{35a_4} + \mathcal{R}_4, \qquad a_3 H^2 = -\theta p_3 - \frac{\theta^2}{10a_3} + \mathcal{R}_3.$$

Define

$$E = \frac{5183683}{5659500}, \quad V_c = \frac{145858703}{581042000}, \quad J_c = \frac{7989533}{14526050}, \quad \mathcal{R}_0 = \mathcal{R}_3 + \mathcal{R}_4 + \eta r_4 + \theta r_3 \geq 0.$$

Substituting the directional identities and (5.8) into (5.7), then writing $L = \overline{L} - r_{\mathrm{reg}}$, gives the exact identity

$$D = d_0 + Ee + V_c V + J_c J - \ell\overline{L} + \mathcal{R}_0 + r_{\mathrm{rad}} + \ell r_{\mathrm{reg}}. \tag{6.3}$$

Here

$$d_0 = \delta_{\mathrm{q}} + \eta\ell_4 + \theta\ell_3 - \frac{2\eta^2}{35a_4} - \frac{\theta^2}{10a_3},$$

which simplifies to (5.1). The coefficient checks are

$$E = b_e - \eta/15 + 9\theta/40, \quad V_c = b_V - 19\eta/1400 - \theta/8 + 3/50, \quad J_c = b_J - 27/50.$$

All the remainders in this identity refer to the same probability law.

Put $B_c = V_c - 5\ell/18$ and $C_c = J_c - \ell/4$. The exact rational values are

$$B_c = \frac{42565069}{653672250}, \qquad C_c = \frac{88944217}{232416800}.$$

For $u = \sqrt{e}$, $z = \sqrt{V}$, and $j = \sqrt{J}$, the remaining quadratic has the explicit completion

$$\begin{aligned} Ee + V_c V + J_c J - \ell\overline{L} = {} & \frac{13}{200}\left(z - \frac{937}{273}qu\right)^2 + \frac{19}{50}\left(j - \frac{2811}{2128}qu\right)^2 \\ & + \left(\frac{183}{200} - \frac{725029049}{278853120}q^2\right)e \\ & + \frac{10481}{11319000}e + \frac{305491}{2614689000}V + \frac{625833}{232416800}J. \end{aligned} \tag{6.4}$$

Indeed the last three fractions are $E - 183/200$, $B_c - 13/200$, and $C_c - 19/50$. Also

$$\frac{725029049}{278853120} = \frac{7\ell}{4} + \frac{\ell^2}{9(13/200)} + \frac{9\ell^2}{16(19/50)}.$$

By (6.1), the coefficient of e in parentheses is at least

$$\frac{183}{200} - \frac{20}{57}\frac{725029049}{278853120} = \frac{53754367}{19868284800} > 0.$$

Thus every term on the right of (6.3) after d_0 is nonnegative. The displayed squares give an explicit certificate for the remaining quadratic form.

Proposition 6.1. *Every isotropic law on $\mathbb{R}^3$ whose radii lie almost surely in a positive interval of endpoint ratio at most 3 satisfies $D \geq d_0$. In the branch $D < 1/20$, (6.3)–(6.4) give the full common-remainder certificate.*

Proof. The small branch was proved above. Otherwise $D \geq 1/20 > d_0$. For explicit exact comparisons use $682/305 < \sqrt{5} < 161/72$ and $433/250 < \sqrt{3} < 97/56$, each verified by squaring. They yield

$$d_0 - 7/144 > \frac{940735364069}{29456638041600000} > 0, \qquad 1/20 - d_0 > \frac{620701786231}{482895705600000} > 0.$$

No $B = 0$ division occurs: it is excluded by $e < 1/20$ in the small branch. □

6.2 A reserve in the quartic slack

Throughout this section assume $0 < r_- \le R \le r_+ < \infty$ almost surely, with $r_+/r_- \le \sqrt{8}$. Write $g(w) = (1 - 9w^2)/25$. Then

$$\frac{(R - R')^2}{RR'} \le \sqrt{8} + \frac{1}{\sqrt{8}} - 2 = \frac{9\sqrt{2}}{4} - 2 \le \frac{32}{27}.$$

For the last inequality, $\sqrt{2} < 344/243$ follows by squaring, with $(344/243)^2 - 2 = 238/59049 > 0$. Put $z_{\mathrm{rad}} = 3(R - R')^2/(4RR')$, so $0 \le z_{\mathrm{rad}} \le 8/9$. The actual remainder in (5.8) now has the exact decomposition

$$r_{\mathrm{rad}} = \frac{P}{9} + r_{\mathrm{tr}}, \qquad r_{\mathrm{tr}} = \mathbb{E}\big[RR' \{(8/9 - z_{\mathrm{rad}})\sigma(w) + z_{\mathrm{rad}}(\sigma(w) - g(w))\}\big] \ge 0. \tag{6.5}$$

This uses both pointwise inequalities of Lemma 5.1, including where $g < 0$. It is not a second independent budget or a new copy of P.

6.2.1 A contact polynomial without a cubic term

Using the same interpolation nodes and tensor weights, set

$$h_*(t) = (t - 1)(t - 1/2)(t + 3/8), \qquad z_*(t) = (t + 9/8)h_*(t) = t^4 - \frac{85}{64}t^2 + \frac{15}{128}t + \frac{27}{128}.$$

The factor $t + 9/8$ cancels the cubic term. We need the following elementary bound, for which we give an explicit proof.

Lemma 6.2. *For $-1 \le t \le 1$, $z_*(t)^2 \le 3\sigma(t)/2$.*

Proof. For $0 \le t \le 1$, cancel common factors away from their zeros to obtain

$$\frac{z_*(t)^2}{\sigma(t)} = \frac{41503}{4(8352 + 2560t)} q_*(t), \qquad q_*(t) = (1 - t)(t + 3/8)^2(t + 9/8)^2.$$

The cancelled expressions extend continuously at the zeros. Its derivative is

$$q_*'(t) = (t + 3/8)(t + 9/8)(165/64 - t/2 - 5t^2).$$

The last factor decreases on $[0, 1]$, equals $13/576 > 0$ at $2/3$, and equals $-33/256 < 0$ at $11/16$. Thus the maximum is within $[2/3, 11/16]$. On that interval $(t + 3/8)(t + 9/8) < 2$ and $q_*'(t) \le 13/288$. The interval has length $1/48$, so

$$q_*(t) \le \frac{1155625}{995328} + \frac{13}{13824} < \frac{6}{5};$$

the final positive difference is $189163/4976640$. Consequently $z_*^2/\sigma \le 41503/27840 < 3/2$ on this half interval. For $t = -x$, $0 \le x \le 1$, the other factorization gives

$$\frac{z_*(-x)^2}{\sigma(-x)} = \frac{41503\, p_*(x)^2}{64(928 + 82x - 160x^2)}, \qquad p_*(x) = (1 + x)(x + 1/2)(9/8 - x) \ge 0.$$

The quadratic denominator factor is at least 850, by concavity and its endpoint values. With $y = x - 1/2 \in [-1/2, 1/2]$,

$$p_*(x) = 15/16 + y/16 - 15y^2/8 - y^3 \le 15/16 + y/16 - 11y^2/8 = \frac{1321}{1408} - \frac{11}{8}(y - 1/44)^2 < 1.$$

It follows that $z_*^2/\sigma \le 41503/54400 < 3/2$ here. At every cancelled zero the claimed inequality follows directly or by continuity. These two intervals cover the domain. □

Remark 6.3. The constant $3/2$ is not sharp: a dense grid scan gives $\sup_{[0,1]} z_*^2/\sigma = 1.200224\ldots$ and $\sup_{[-1,0]} z_*^2/\sigma = 0.615384\ldots$. We keep $3/2$ because it keeps every subsequent coefficient rational and hand-verifiable. The loss enters the certificate only through r_{ct} and through the divisor 125 in (6.8); both are quantified in Appendix A.

In particular the following displayed square proves a useful linear payment:

$$\sigma(t) - \frac{z_*(t)}{125} + \frac{3}{125000} = \left(\sigma(t) - \frac{2}{3}z_*(t)^2\right) + \frac{2}{3}\left(z_*(t) - \frac{3}{500}\right)^2 \ge 0. \tag{6.6}$$

All these functions are bounded on $[-1, 1]$. Integrating against the actual nonnegative weight RR' uses only $\mathbb{E}R < \infty$, not a fourth or eighth radial moment. The actual contact moment is

$$Z_* := \mathbb{E}[RR' z_*(U \cdot U')] = K^2 + \frac{7}{320} - \frac{15}{128}e - \frac{561}{4480}V + \frac{211}{448}J. \tag{6.7}$$

Indeed the RR' moments of $1, w, w^2, w^4$ are respectively $3 - V, 1 - e, 1 - J, K^2 + 6\|Z\|^2/7 + (3 - V)/5$, all for this same law. Define

$$\mathcal{C} = Z_*/125 - 3C^2/125000, \qquad r_{\mathrm{ct}} = P - \mathcal{C} \ge 0. \tag{6.8}$$

The scalar $\mathcal{C}$ need not itself be nonnegative. Its exact contribution is

$$\frac{\mathcal{C}}{9} = \frac{103}{9000000} + \frac{K^2}{1125} - \frac{e}{9600} + \left(-\frac{187}{1680000} + \frac{1}{375000}\right)V + \frac{211}{504000}J. \tag{6.9}$$

6.2.2 The full joint certificate

For $D < 1/20$ define the explicitly nonnegative quadratic expression

$$\begin{aligned} Q = {} & \frac{13}{200}\left(\sqrt{V} - \frac{937}{273}q\sqrt{e}\right)^2 + \frac{19}{50}\left(\sqrt{J} - \frac{2811}{2128}q\sqrt{e}\right)^2 \\ & + \left(\frac{183}{200} - \frac{725029049}{278853120}q^2\right)e \\ & + \frac{148831}{181104000}e + \frac{42847433}{5229378000000}V + \frac{65082103}{20917512000}J. \end{aligned} \tag{6.10}$$

The parenthetical coefficient is positive by (6.1) and the same exact lower bound following (6.4). The last coefficients are obtained from those in (6.4), respectively by adding $-1/9600$, $-187/1680000 + 1/375000$, and $211/504000$. Substitution of (6.5)–(6.9) into (6.3) yields the

exact identity

$$D = d_1 + Q + \mathcal{R}_0 + \ell r_{\mathrm{reg}} + \frac{K^2}{1125} + r_{\mathrm{tr}} + \frac{r_{\mathrm{ct}}}{9}. \tag{6.11}$$

Every remainder is defined from the same X, its independent copy and its complete Gram matrix.

Proposition 6.4. *Every isotropic law whose radii lie almost surely in a positive bounded interval of endpoint ratio at most $\sqrt{8}$ satisfies $D \geq d_1$. In the branch $D < 1/20$,* (6.11) *retains all displayed remainders.*

Proof. In the small branch all terms after d_1 are nonnegative. In the other branch $D \geq 1/20 > d_1$. To verify the last comparison entirely by rationals, use $\sqrt{5} < 161/72$ and $\sqrt{3} > 433/250$ in (5.1); they give

$$\frac{1}{20} - d_1 > \frac{1025292188593}{804826176000000} > 0.$$

No division by $B = 0$ occurs in the small branch, and the other branch requires none. The radial hypothesis will be verified for a maximizing law in Section 6.3, rather than imposed on every isotropic law. □

6.3 Transfer to graphs and proof of the main theorem

We return to a maximizing graphon. Put

$$D_* = (3 + \sqrt{5})/2 - 2\mathcal{M}.$$

By the threshold identity in Lemma 3.6, its canonical law $X = F$ satisfies $A + B = 2\mathcal{M}$, so the deficit D of the preceding sections equals D_*.

Lemma 6.5. *If W is an actual maximizer and $D_* = (3 + \sqrt{5})/2 - 2\mathcal{M} < 1/20$, then $\lambda_3 > 44/375$, $\lambda_1 < 15/16$, and the canonical frame has positive bounded radii of endpoint ratio less than $\sqrt{8}$.*

Proof. The small branch and $\sqrt{5} > 682/305 > 559/250$ give

$$\mathcal{M} > \frac{3 + \sqrt{5}}{4} - \frac{1}{40} > \frac{321}{250}.$$

Lemma 3.1 gives $\lambda_3 = \mathcal{M} - \Sigma_2(W) > 321/250 - 7/6 = 44/375$. For $a = \lambda_1$, the Hilbert–Schmidt trace and the constant-function trial give

$$\lambda_1^2 + \lambda_2^2 + \lambda_3^2 \leq \iint W^2 \leq \iint W \leq \lambda_1, \qquad \mathcal{M} \leq a + \sqrt{2a(1-a)}, \quad 0 < a \leq 1.$$

The function on the right is decreasing on $[15/16, 1]$; differentiation on the interior reduces this to $(2a-1)^2 > 2a(1-a)$ there. Moreover

$$\frac{15}{16} + \sqrt{\frac{15}{128}} < \frac{321}{250}, \qquad \left(\frac{321}{250} - \frac{15}{16}\right)^2 - \frac{15}{128} = \frac{11499}{4000000} > 0.$$

Thus $a < 15/16$. By the actual ellipsoid identity,

$$\sqrt{\mathcal{M}/\lambda_1} \le R \le \sqrt{\mathcal{M}/\lambda_3}, \qquad \frac{\operatorname{ess\,sup} R}{\operatorname{ess\,inf} R} \le \sqrt{\lambda_1/\lambda_3} < \sqrt{5625/704} < \sqrt{8},$$

where $8 - 5625/704 = 7/704$. No simplicity assumption was made on the positive eigenvalues; the argument uses the whole orthonormal frame. □

Remark 6.6. Lemma 5.2 is stated with endpoint ratio 3, whereas the present lemma supplies $\sqrt{8} < 3$ for an actual maximizer; the two hypotheses are therefore compatible, with slack. Using $\sqrt{8}$ in Lemma 5.2 would replace the factor $4/3$ in (5.8) by $32/27$, a 12.5% gain on P, which is not needed here.

6.3.1 Positive-mass self-pairs

By Theorem 1.2, write the canonical law as $\sum_{i=1}^{m} p_i \delta_{v_i}$ with $m \le 24$ and all $p_i > 0$. Set $R_i = |v_i|$, $u_i = v_i/R_i$, $w_{ij} = u_i \cdot u_j$, and

$$\kappa = \frac{3}{125000}, \qquad h(t) = \sigma(t) - \frac{z_*(t)}{125} + \kappa, \qquad \chi = \sum_i p_i^2 R_i^2.$$

Equation (6.6) gives $h \ge 0$ and $h(1) = \kappa$. The same contact remainder in (6.8) therefore has the exact split

$$r_{\mathrm{ct}} = \kappa\chi + r_{\mathrm{off}}, \qquad r_{\mathrm{off}} = \sum_{i \ne j} p_i p_j R_i R_j h(w_{ij}) \ge 0. \tag{6.12}$$

These self-pairs have positive product mass. They are not values assigned to the null diagonal $x = y$ of the underlying nonatomic space. The estimate $\chi \ge C^2/m$ is the usual Cauchy–Schwarz cardinality bound. For related weighted-energy arguments, see [1, Theorem 2.1 and the subsequent discussion].

In the branch $D_* < 1/20$, let

$$\widehat{Q} = Q - \frac{V}{9000000}.$$

It is nonnegative: in (6.10) only the last separate V coefficient changes, to

$$\frac{42847433}{5229378000000} - \frac{1}{9000000} = \frac{14088797}{1743126000000} > 0.$$

Proposition 6.7. *For every maximizing graphon with $D_* < 1/20$,*

$$\begin{aligned} D_* = d + \widehat{Q} + \mathcal{R}_0 + \ell r_{\mathrm{reg}} + \frac{K^2}{1125} + r_{\mathrm{tr}} \\ + \frac{\kappa}{9}\left(\chi - \frac{C^2}{24}\right) + \frac{r_{\mathrm{off}}}{9}. \end{aligned} \tag{6.13}$$

Every term after d is nonnegative. Consequently $\mathcal{M} \le c$.

Proof. Lemma 6.5 verifies the radial hypothesis of Proposition 6.4. Substitute (6.12) once

into (6.11). Since $C^2 = 3 - V$,

$$\frac{\kappa\chi}{9} = \frac{1}{3000000} - \frac{V}{9000000} + \frac{\kappa}{9}\left(\chi - \frac{C^2}{24}\right).$$

This gives (6.13). Moreover

$$\chi - \frac{C^2}{24} = \sum_i \left(p_i R_i - \frac{C}{m}\right)^2 + C^2\left(\frac{1}{m} - \frac{1}{24}\right) \geq 0.$$

No lower bound on an individual mass is needed. The complementary branch also gives $D_* \geq d$, since the rational estimates used above imply

$$\frac{1}{20} - d > \frac{1025023913201}{804826176000000} > 0.$$

Thus $D_* \geq d$ in both branches, or equivalently $\mathcal{M} \leq c$. □

The atomic term changes the coefficient by $1/6000000$ relative to $(3+\sqrt{5})/4 - d_1/2$. Most of the improvement in Theorem 1.1 comes from the preceding joint moment estimate, not from this last term.

6.3.2 All isotropic laws and the exact finite shift

Proof of Theorem 1.1. Any Borel law on $\mathbb{R}^3$ can be realized on a standard nonatomic probability space, by adjoining an independent uniform variable if necessary. For its isotropic X, the coordinates are an orthonormal three-frame and $W_X = \mathbf{1}_{\{X\cdot X' > 0\}}$ is a measurable symmetric binary kernel. Therefore

$$\frac{\mathbb{E}|X \cdot X'| + |\mathbb{E}X|^2}{2} = \iint W_X X \cdot X' \leq \Sigma_3(W_X) \leq \mathcal{M} \leq c.$$

This proves (1.10), including zero vectors and unbounded radii. The radial restriction was used only for a maximizing canonical law.

For a finite graph, the exact-rank Ky Fan formula reads

$$\Lambda_3(G) = \max_{Q = Q^T = Q^2,\ \operatorname{tr} Q = 3} \operatorname{tr}(A_G Q).$$

Factor $Q = YY^T$ with orthonormal columns, let y_i be its rows, and let X be uniform on $\sqrt{n}\, y_1, \ldots, \sqrt{n}\, y_n$. Then

$$\mathbb{E}XX^T = I_3, \qquad \mathbb{E}|X \cdot X'| = \|Q\|_1/n, \qquad |\mathbb{E}X|^2 = \mathbf{1}^T Q \mathbf{1}/n,$$

where $\|Q\|_1$ denotes the entrywise absolute sum. Since $Q_{ii} \geq 0$ and $\sum_i Q_{ii} = 3$, the zero adjacency diagonal gives

$$\operatorname{tr}(A_G Q) \leq \sum_{i \neq j} (Q_{ij})_+ = \frac{\|Q\|_1 + \mathbf{1}^T Q \mathbf{1}}{2} - 3 \leq cn - 3.$$

Taking the maximum proves (1.9). No positivity assumption on $\lambda_3(G)$ is used; repeated eigenvalues, zero rows and $n = 3$ cause no exception. □

6.4 Exact scope of the bound

The transfer argument above uses two properties of A_G only: its diagonal is zero, and its off-diagonal entries lie in $[0, 1]$. Both are shared by a larger class, and the bound therefore holds there without further work. The scope of such a statement is easy to mis-state, so we record it precisely, and we record at the same time that no novelty is claimed for it: the same observation underlies the proof of [16, Proposition 6.1], and the matrix setting is explicit in the title and content of [12].

Proposition 6.8 (Weighted form)**.** *Let $n \geq 3$ and let $A \in \mathbb{R}^{n\times n}$ be symmetric with $A_{ii} = 0$ and $0 \leq A_{ij} \leq 1$ for $i \neq j$. Then*

$$\lambda_1(A) + \lambda_2(A) + \lambda_3(A) \leq cn - 3,$$

with the constant c of (1.8)*.*

Proof. Apply Lemma 2.3 with $r = 3$. The isotropic law produced there is built from the rows of a rank-three projection and satisfies no radial hypothesis, so Theorem 1.1 applies to it and gives $\Gamma_3 \leq c$. The constant term is $-r = -3$ and is unchanged. □

Remark 6.9. Proposition 6.8 covers $[0, 1]$-weighted loopless graphs, hence also every graph with edge weights in $[0, 1]$ arising from a discretised graphon. It fails for weights above 1 and for multiple edges: if $A = 2A_G$ then $\Lambda_3(A) = 2\Lambda_3(G)$, which exceeds $cn - 3$ for large n whenever $\Lambda_3(G)/n > c/2$. The numerical tests of Section 7.4 are consistent with this.

Weights cannot improve the asymptotic constant either, so the extremal problem does not become easier by relaxation.

Proposition 6.10 (Weights give no asymptotic gain)**.** *Let $\mathcal{M}_{\mathrm{w}}(n)$ be the supremum of $\Lambda_3(A)$ over symmetric $A \in \mathbb{R}^{n\times n}$ with $A_{ii} = 0$ and $0 \leq A_{ij} \leq 1$. Then*

$$\frac{\mathcal{M}_{\mathrm{w}}(n)}{n} \longrightarrow \mathcal{M} \qquad (n \to \infty),$$

the same limit as for unweighted graphs.

Proof. Every such A defines a step graphon W_A with $T_{W_A} = A/n$, so by (2.6) $\Lambda_3(A) \leq n\Sigma_3(W_A) \leq n\mathcal{M}$; hence $\limsup_n \mathcal{M}_{\mathrm{w}}(n)/n \leq \mathcal{M}$. For the reverse inequality, $\mathcal{M}$ is attained by some graphon W by Lemma 3.2. By the compactness and density statements of Section 2.2 there are graphs G_m with $\delta_\square(W_{G_m}, W) \to 0$, and Σ_3 is continuous in $\delta_\square$ because $|\Sigma_3(W_1) - \Sigma_3(W_2)| \leq 3\|T_D\|_{2\to 2} \leq 3(4\|D\|_\square)^{1/3}$ by (2.7). Writing $n_m = |V(G_m)|$, the identity $\Lambda_3(G_m) = n_m\Sigma_3(W_{G_m}) + \sum_{i\leq 3} \min(\lambda_i(G_m), 0)$ and the bound $\lambda_i(G_m) \geq -(n_m - 1)$ give

$$\frac{\mathcal{M}_{\mathrm{w}}(n_m)}{n_m} \geq \frac{\Lambda_3(G_m)}{n_m} \geq \Sigma_3(W_{G_m}) - 3\,\frac{n_m - 1}{n_m} \longrightarrow \mathcal{M}.$$

Replacing G_m by a blow-up if necessary, which does not change its graphon, we may assume $n_m \to \infty$. Since $\mathcal{M}_{\mathrm{w}}(n)/n \le \mathcal{M}$ for every n, the limit exists and equals $\mathcal{M}$. □

Remark 6.11. The padding term is nonnegative: by (2.6),

$$n\Sigma_3(W_G) - \Lambda_3(G) = -\sum_{i\le 3} \min(\lambda_i(G), 0) \ge 0.$$

It is therefore harmless in the direction used above, where only a lower bound for $\mathcal{M}_{\mathrm{w}}(n)$ is needed.

Chapter 7

Numerical verification and scope

This chapter reports computations. They play no role in any proof: every theorem in this thesis is proved from the axioms, and the experiments are recorded because they locate the extremal graphs, because they quantify how much slack the bound has at small orders, and because they delimit the class of matrices to which the bound extends. Each table is labelled *numerical* in the sense of Section 1.6, except where an exact identity is stated and proved alongside it.

7.1 Protocol

Three kinds of computation were run, with deliberately separated roles.

Exact arithmetic. Every displayed identity, factorisation, completion of squares and positivity assertion in Chapters 5 and 6 was re-checked with exact rationals and with elements of the quadratic fields $\mathbb{Q}(\sqrt{3})$ and $\mathbb{Q}(\sqrt{5})$ represented as pairs of rationals, never in floating point. The constants c, c_{w}, $(3+\sqrt{5})/4$ and $3/\sqrt{5}$ were recomputed to sixty decimal places:

$$\begin{aligned} c &= 1.28467325572391850506073278586458926092723839129029128018 2\ldots,\\ c_{\mathrm{w}} &= 1.28156819687216787675243501399191724068175671228890035389 0\ldots,\\ (3+\sqrt{5})/4 &= 1.30901699437494742410229341718281905886015458990288143106 7\ldots,\\ 3/\sqrt{5} &= 1.34164078649987381784550420123876574126437101576691543456 2\ldots, \end{aligned}$$

from which $c - c_{\mathrm{w}} = 0.00310505885175062830829777187267202024548 16\ldots$ and $(3+\sqrt{5})/4 - c = 0.02434373865102891904156063131822979793291 6\ldots$. The six-part quotient value at the optimising weight was confirmed equal to c_{w} to forty digits.

Exhaustive enumeration. For $3 \le n \le 7$ all $2^{n(n-1)/2}$ labelled graphs were enumerated, Λ_3 was computed with a symmetric eigensolver in double precision, and the result was compared with $cn-3$ evaluated from the exact expression for c. A 7×7 symmetric eigenvalue computation in double precision has backward error of order 10^{-15}, while the smallest margin encountered is 2.2×10^{-2}; the conclusions are therefore insensitive to rounding.

Randomised sampling. For the weighted tests, symmetric matrices with zero diagonal and independent uniform $[0,1]$ upper-triangular entries were sampled with a fixed seed, so the

runs are reproducible.

7.2 Exhaustive verification for $n \leq 7$

The total number of labelled graphs examined is $8 + 64 + 1024 + 32768 + 2097152 = 2\,131\,016$. There were no violations. The extremal values, the extremal graphs up to isomorphism and the number of labelled copies are in Table 7.1. At $n = 3$ all eight graphs tie, since $\Lambda_3 = \operatorname{tr} A_G = 0$ there; at $n = 4$ the extremal graph is unique; at $n = 5$ there are exactly two extremal isomorphism classes; at $n = 6$ and $n = 7$ the extremal graph is again unique.

Table 7.1: Exhaustive enumeration over all labelled graphs, $3 \leq n \leq 7$. "Margin" is $cn - 3 - \max \Lambda_3$; the exact value of $\max \Lambda_3$ is printed where it is known in radicals. At $n = 3$ every graph satisfies $\Lambda_3 = \operatorname{tr} A_G = 0$.

n	# graphs	$\max \Lambda_3$	$cn - 3$	margin	extremal graph
3	8	0	0.854019767	0.854019767	—
4	64	2	2.138693023	0.138693023	C_4
5	1024	$1 + \sqrt{5} = 3.236067977$	3.423366279	0.187298301	C_5, $K_1 \vee C_4$
6	32768	$\sqrt{5} + \sqrt{6} = 4.685557720$	4.708039534	0.022481814	$K_1 \vee C_5$
7	2097152	$\frac{1+\sqrt{41}}{2} + \sqrt{5} = 5.937630096$	5.992712790	0.055082694	$K_2 \vee C_5$

The number of labelled maximisers is 3 at $n = 4$, 27 at $n = 5$, 72 at $n = 6$ and 252 at $n = 7$. The value $72 = 6!/10$ matches $|\operatorname{Aut}(K_1 \vee C_5)| = 10$, and $252 = 7!/20$ matches $|\operatorname{Aut}(K_2 \vee C_5)| = 2 \cdot 10 = 20$; for $n = 7$ all 252 maximisers were additionally checked to have degree sequence $(6, 6, 4, 4, 4, 4, 4)$ and to induce a 5-cycle on the five vertices of degree four. The split $27 = 12 + 15$ at $n = 5$ matches $5!/10 = 12$ labelled copies of C_5 and 15 labelled copies of $K_1 \vee C_4$; the two classes are not isomorphic but are exactly tied, which is why uniqueness fails there.

Two features of Table 7.1 deserve comment. First, the order $n = 3$ is the smallest for which the bound is even meaningful: for $n = 1, 2$ the right-hand side $cn - 3$ is negative while $\Lambda_3 = \operatorname{tr} A_G = 0$ whenever $n \leq 3$, so the hypothesis $n \geq 3$ in Theorem 1.1 cannot be dropped. Second, the margin is smallest at $n = 6$, which is exactly the order of the template $K_1 \vee C_5$ that produces c_{w}; the bound is therefore closest to being tight precisely where the lower-bound construction lives, which is the numerical counterpart of the gap $c - c_{\mathrm{w}}$.

7.3 Blow-up families

The template $H = K_1 \vee C_5$ generates an explicit infinite family whose Λ_3 can be computed in closed form.

Proposition 7.1 (Equal blow-ups of $K_1 \vee C_5$)**.** *Let $q \geq 1$ and let $G = H[q]$ be the graph on $n = 6q$ vertices obtained by replacing every vertex of H by a clique of size q and every edge of H by a complete bipartite graph. Then*

$$\Lambda_3(G) = \frac{\sqrt{5} + \sqrt{6} + 3}{6}\, n - 3, \qquad \frac{\sqrt{5} + \sqrt{6} + 3}{6} = 1.2809262867\ldots.$$

Proof. By Lemma 2.1 the spectrum of G consists of $q\lambda + q - 1$ for $\lambda \in \operatorname{spec} H$ together with -1 repeated $6(q-1)$ times. The spectrum of H is $1+\sqrt{6}$, $\frac{\sqrt{5}-1}{2}$ twice, $1-\sqrt{6}$ and $-\frac{1+\sqrt{5}}{2}$ twice, so the three largest eigenvalues of G are $q(1+\sqrt{6})+q-1$ and $q\frac{\sqrt{5}-1}{2}+q-1$ twice; for $q \geq 1$ these exceed both $q(1-\sqrt{6})+q-1$ and -1. Their sum is $q(\sqrt{6}+\sqrt{5}+3)-3$, which equals the claimed expression at $n = 6q$. □

The coefficient $1.2809262867\ldots$ is strictly below both c_{w} and c. Equal part sizes are therefore suboptimal, and the loss is quantified:

$$c_{\mathrm{w}} - \frac{\sqrt{5}+\sqrt{6}+3}{6} = 0.0006419102\ldots, \qquad c - \frac{\sqrt{5}+\sqrt{6}+3}{6} = 0.0037469690\ldots.$$

Table 7.2 confirms the closed form numerically and shows the distance to both comparison lines.

Table 7.2: Equal blow-ups $G = H[q]$ of $H = K_1 \vee C_5$, $n = 6q$. All values numerical; the column Λ_3 agrees with the closed form of Proposition 7.1 to the printed precision.

q	n	$\Lambda_3(G)$	Λ_3/n	$\Lambda_3 - (cn-3)$	$\Lambda_3 - (c_{\mathrm{w}}n-3)$
1	6	4.685558	0.780926	−0.022482	−0.003851
2	12	12.371115	1.030926	−0.044964	−0.007703
3	18	20.056673	1.114260	−0.067445	−0.011554
5	30	35.427789	1.180926	−0.112409	−0.019257
10	60	73.855577	1.230926	−0.224818	−0.038515
20	120	150.711154	1.255926	−0.449636	−0.077029
50	300	381.277886	1.270926	−1.124091	−0.192573
100	600	765.555772	1.275926	−2.248181	−0.385146
250	1500	1918.389430	1.278926	−5.620454	−0.962865
500	3000	3839.778860	1.279926	−11.240907	−1.925730

The optimal part sizes are given by Lemma 3.2: apex mass $p^* = (91-25\sqrt{7})/126 = 0.1972715653\ldots$ and rim mass $b^* = (7+5\sqrt{7})/126 = 0.1605456869\ldots$ for each of the five rim parts. Realising these as integers introduces a rounding error of order one vertex per part, so the resulting graphs fall short of $c_{\mathrm{w}}n - 3$ by an amount that decreases with n. Table 7.3 records this convergence, using the general quotient form (2.8).

Table 7.3: Optimally weighted blow-ups of $K_1 \vee C_5$: one apex clique and five rim cliques, sizes rounded from $(p^*, b^*, \ldots, b^*)n$. The last column tends to 0, which is the numerical form of $\Sigma_3(S) = c_{\mathrm{w}}$ for the optimal six-part step graphon S.

n	part sizes	Λ_3	Λ_3/n	$\Lambda_3 - (cn-3)$	$\Lambda_3 - (c_{\mathrm{w}}n-3)$
600	$(120, 96^5)$	765.937967	1.276563	−1.865986	−0.002951
1200	$(235, 193^5)$	1534.880189	1.279067	−3.727718	−0.001647
3000	$(590, 482^5)$	3841.703863	1.280568	−9.315904	−0.000728

The two tables together exhibit the structure of the gap. The optimally weighted family approaches $c_{\mathrm{w}}n-3$ from below and is far from $cn-3$, while the bound $cn-3$ is not attained by any graph found here. Nothing in this chapter decides whether a different family could exceed

$c_{\mathrm{w}}n - 3$; by Proposition 6.10 no family of weighted graphs can exceed $\mathcal{M}n - 3$ asymptotically, and $\mathcal{M}$ itself is only bracketed.

7.4 Weighted matrices

Proposition 6.8 extends the bound to every symmetric matrix with zero diagonal and off-diagonal entries in $[0, 1]$. Table 7.4 reports randomised tests of that statement. No violation occurred. A control run of 200 matrices of order 30 with entries uniform in $[0, 2]$ produced 198 violations of $\Lambda_3 \le cn - 3$, so the range restriction $A_{ij} \le 1$ is the operative hypothesis and not an artefact of the sampling; this is the numerical counterpart of Remark 6.9.

Table 7.4: Random $[0, 1]$-weighted loopless matrices, 4000 samples per order, fixed seed. "Max" is the largest Λ_3 observed; "bound" is $cn - 3$. No violation occurred at any order.

n	samples	violations	max Λ_3	$cn - 3$
4	4000	0	1.663987	2.138693
6	4000	0	3.637893	4.708040
9	4000	0	6.244673	8.562059
15	4000	0	10.708949	16.270099
30	4000	0	20.579009	35.540198
60	4000	0	38.337336	74.080395

The random weighted matrices are far from extremal: at $n = 60$ the observed maximum is barely half the bound. This is expected, because a random weighted matrix has a dense, near-semicircular spectrum whose third eigenvalue is small compared with n, whereas the extremal constructions are sparse-template blow-ups with three large eigenvalues. The experiment is therefore a test of the hypothesis range, not a search for extremals.

7.5 What the computations do not show

Three limitations are stated explicitly so that no reader mistakes numerical evidence for a theorem.

- Exhaustive verification up to $n = 7$ says nothing about $n \ge 8$; the margin at $n = 6$ is small enough that no extrapolation from the table is possible.
- The absence of violations in Table 7.4 does not prove Proposition 6.8; that proposition is proved in Chapter 6 by Lemma 2.3, and the sampling only checks that the proof's hypotheses were not mis-stated.
- The blow-up tables identify a candidate extremal family but do not establish that $\mathcal{M} = c_{\mathrm{w}}$. Proposition 6.10 shows that the asymptotic constant is the same for weighted and unweighted problems; it does not evaluate it.

7.6 Reproducibility

The scripts that produced Tables 7.1 to 7.4 and the exact-arithmetic checks of Section 7.1 are stored in the accompanying repository under `thesis_main/scripts/`, with the raw outputs as JSON in the same directory. Each script is deterministic: the enumeration is exhaustive, and the randomised tests use the fixed seed 20261104. The exact-arithmetic checks use only integer and quadratic-field operations and print their results at sixty digits, so they can be reproduced without any numerical library.

Chapter 8

Conclusion and open problems

8.1 Summary of what is proved

The thesis establishes four results, listed here in the order in which they are used rather than in order of importance.

1. **Attainment and two-sided bracketing** (Lemma 3.2). The variational quantity $\mathcal{M} = \sup_W \Sigma_3(W)$ is attained, every maximiser has three positive leading eigenvalues, and $c_{\mathrm{w}} \le \mathcal{M} \le (1+\sqrt{3})/2$ with $c_{\mathrm{w}} = (9+\sqrt{5}+2\sqrt{35})/18$ produced by an explicit six-part step graphon modelled on $K_1 \vee C_5$.

2. **Finite canonical support** (Theorem 1.2). The canonical law of *every* maximiser is a convex combination of at most 24 points of the ellipsoid $\{v^T L v = \mathcal{M}\}$; the maximiser is almost everywhere the binary threshold kernel determined by the signs of the pairwise dot products; the support points satisfy the exact algebraic system $\sum_i p_i v_i v_i^T = I_3$, $v_i^T L v_i = \mathcal{M}$, $\sum_j p_j B_{ij} v_j = L v_i$; and every positive clique has at most six linearly independent types.

3. **Joint moment certificate** (Theorem 1.1). Every Borel probability law on $\mathbb{R}^3$ with $\mathbb{E}XX^T = I_3$ satisfies $\mathbb{E}|X \cdot X'| + |\mathbb{E}X|^2 \le 2c$ with $c = 1.284673255723918505\ldots$. The proof is an exact identity in which the mean deficit, the radial variance, the regression error, the two harmonic moments, the quartic slack and the atomic self-pair term all appear with explicit nonnegative coefficients.

4. **Finite bound and its scope** (Theorem 1.1, Propositions 6.8 and 6.10). Every graph of order $n \ge 3$ satisfies $\Lambda_3(G) \le cn - 3$; the same inequality holds for every symmetric matrix with zero diagonal and off-diagonal entries in $[0,1]$; and passing from graphs to such matrices does not change the asymptotic extremal constant.

The improvement over [16, Proposition 6.1] is a coefficient decrease of $0.02434374\ldots$ with the constant term -3 unchanged; the improvement over [5, Theorem 3.4.1] is $3/\sqrt{5} - c = 0.05696753\ldots$. Both statements are proved, not numerical, and both constants are printed to sixty digits in Section 7.1.

8.2 The residual gap

The explicit construction in Lemma 3.2 leaves the interval

$$\frac{9+\sqrt{5}+2\sqrt{35}}{18} \le \mathcal{M} \le 1.284673255723918505\ldots.$$

The lower endpoint comes from a weighted wheel, but its global optimality is not established here. Its provenance is the subject of Remark 3.3: the same constant and the same optimising weight were announced earlier, conditionally, in [13, 14], whose certificate argument would close the interval at the lower endpoint were it to survive the independent audit that its author requests. Independently of that outcome, the unconditional question left open here is the gap

$$c - c_{\mathrm{w}} = 0.003105058851750628\ldots,$$

and removing it is the natural next problem. The support bound reduces the structure of every maximizer to finitely many positive-mass types without classifying the possible templates. It does not imply that the binary matrix B is nonsingular, that every maximizer has six types, or that individual masses are uniformly bounded away from zero. Determining the optimum remains a separate problem.

The exact identity (6.13) also distinguishes a quantitative certificate from a qualitative equality obstruction. Any further decrease obtained from this certificate must bound a remaining term from below under the same constraints, without counting the same quartic remainder twice. The present argument does not use an unquantified stability gap.

8.3 Open problems

The following are stated in decreasing order of how much of the present machinery they would reuse. None is proved here.

1. **Close the gap.** Prove or disprove $\mathcal{M} = c_{\mathrm{w}}$, i.e. remove the gap $c - c_{\mathrm{w}} = 0.00310506\ldots$. The obstruction is identified in Appendix A: the certificate loses through the non-sharp constant $3/2$ of Lemma 6.2 (the true supremum is $1.200224\ldots$ on $[0,1]$), through the divisor 125 in (6.8), and through the endpoint-ratio hypotheses 3 and $\sqrt{8}$ that are stronger than needed. Replacing $3/2$ by a sharp contact constant is the cheapest identified improvement and is a finite exact-arithmetic computation, but it was not carried out.

2. **Determine the joint feasible region.** Let $\Gamma = \overline{\{(\mathbb{E}|X \cdot X'|, |\mathbb{E}X|^2)\}}$ over isotropic laws on $\mathbb{R}^3$. The two coordinate bounds are known and sharp, $\mathbb{E}|X \cdot X'| \le (1+\sqrt{5})/2$ and $|\mathbb{E}X|^2 \le 1$, and Theorem 1.1 bounds their sum. The region Γ itself is not known; it is the natural two-dimensional formulation of the mixed projection constant raised in [16, Section 6.2]. This is the problem for which the present method gives a supporting line rather than the whole boundary.

3. **Classify the extremal templates.** The support theorem bounds the number of types but does not classify them. In particular it is not known whether B is nonsingular, whether every maximiser has exactly six types, or whether the masses are bounded away from zero uniformly over maximisers. The numerical evidence of Chapter 7 points to the family $K_m \vee C_5[\cdot]$ but proves nothing.

4. **Exact finite-order extremal graphs.** Determine $M_3(n)$ and the graphs attaining it for every n. At present this is known only for $n \le 7$ by exhaustive computation (Chapter 7); the extremal graphs there are $K_1 \vee C_5$ for $n = 6$ and $K_2 \vee C_5$ for $n = 7$, which suggests a blow-up family with a congruence-dependent description as in [7], but no conjecture strong enough to state is supported by the data.

5. **Resolve the status of the announced certificate.** The deposits [13, 14] assert $\Lambda_3(G) \le c_{\mathrm{w}} n - 3$ conditionally. Their author requests an independent audit of the admissibility of the archived dual payments, the global sector partition and the exact verifier. Completing that audit would either collapse the interval of Section 8 to a point or produce a counterexample to a published-style claim.

6. **Ring length.** The construction uses C_5. Whether $K_1 \vee C_r$ for $r \ne 5$, or more general joins, produce larger constants has not been investigated here and is recorded as an open computation rather than as a conjecture.

8.4 Adjacent problems

Three neighbouring spectral-sum problems frame the present one and are worth stating precisely, because the methods differ.

The two-term sums. $\Lambda_2 = \lambda_1 + \lambda_2$ is settled: $\Lambda_2(G) \le 8n/7 - 2$ with equality exactly when $7 \mid n$, and the unique extremal graph is the three-part join described in [7], building on [8]. For $\lambda_2 + \lambda_3$ the bound $2n/3 - 2$ follows from the explicit description of the rescaled pair region in [15]. A separate line of the present author's work treated $\lambda_2 + \lambda_3$ with an equality characterisation and a finite-order stability refinement; it was completed independently of and concurrently with [15] and is not reproduced here, since its subject is a different functional.

Weighted two-term sums. For $\lambda_1 + k\lambda_2$ the asymptotic coefficient is already determined for every fixed $k \ge 0$ by the support function of the region in [15]:

$$h(k) = \frac{k + 3 + 2\sqrt{2k^2 - 2k + 4}}{7} \quad (0 \le k \le 3), \qquad h(k) = \frac{k+1}{2} \quad (k \ge 3),$$

so that $\lambda_1 + k\lambda_2 \le h(k)n - (k+1)$ and the coefficient is asymptotically sharp. For $k = 2$ this is $h(2) = (5 + 4\sqrt{2})/7 = 1.5224077499\ldots$. What remains open for $\lambda_1 + 2\lambda_2$ is exactly what remains open for Λ_3: the exact finite-order maximum, the full list of extremal graphs, and a stability theorem. For trees and large order, the extremal tree is known [28].

Individual eigenvalues. $\lambda_3(G) \le n/3 - 1$ is sharp by [16], and c_3 is therefore determined; the interesting individual problem is now c_k for $k \notin \{2, 3, 4, 8, 24\}$, where the upper bound

α_k of [16] and the lower bounds of [24] do not meet.

8.5 What would count as a complete solution

For the reader who wants a single criterion: the problem posed in Section 1.1 is solved when one exhibits an explicit constant $c^\dagger$ and an explicit infinite family of graphs with $\Lambda_3(G) = c^\dagger|V(G)| - 3 + o(1)$, together with a proof that $\Lambda_3(G) \le c^\dagger|V(G)| - 3$ for all G. At present c is proved and c_{w} is attained asymptotically by the weighted blow-ups of $K_1 \vee C_5$, but c_{w} is not proved and c is not known to be attained. Closing either half would be a substantial result; closing both at the same value would finish the problem.

Appendix A

Constants: provenance and machine verification

A reader cannot tell from the statements alone which of the many integer coefficients in this paper are forced by an interpolation problem and which were chosen and then verified. We record the distinction.

Uniquely determined. (i) The polynomial f of §6 is the unique polynomial of degree at most four with double contact to $|w| + w$ at $w = -3/8$ and at $w = 1/2$ and a contact at $w = 1$; five conditions determine five coefficients, so once the nodes are fixed the integers in f (including $41503 = 7^3 \cdot 11^2$) are forced. (ii) z_* of §8 is determined by the same nodes together with the requirement that the cubic term vanish, i.e. by the factor $(t + 9/8)$. (iii) c_{w} of Lemma 3.2 is the value of an explicit one-variable function at the critical point $b = (7+5\sqrt{7})/126$. (iv) The four rational certificates of §7 and §9 are obtained by substituting the stated rational bounds for $\sqrt{5}$ and $\sqrt{3}$ into d_0, d_1, d; no search is involved.

Chosen, then verified exactly. (i) The nodes $-3/8$, $1/2$, 1 themselves, and the majorant h_0 taken from [16, Lemma 4.4]. (ii) The parameters $\ell = 937/1400$, $\eta = 1/10$, $\theta = 6/25$ of §7 and the square coefficients of (6.4) and (6.10); these were selected so that all residual coefficients remain positive, and every positivity claim is then checked by exact rational arithmetic. (iii) The divisor 125 and $\kappa = 3/125000$ of (6.8), and the constant $3/2$ of Lemma 6.2; the quantified loss is discussed in Remark 6.3.

Verification protocol. Every displayed identity, factorization, completion of squares and "> 0" assertion was re-checked in exact arithmetic: rationals as exact fractions and $\sqrt{3}, \sqrt{5}$ in the quadratic fields $\mathbb{Q}(\sqrt{3})$, $\mathbb{Q}(\sqrt{5})$, never in floating point. Pointwise inequalities were additionally scanned on a $2 \times 10^6 + 1$ point grid, and the finite-graph transfer was checked on all 33864 graphs of order $3 \le n \le 6$. The verification scripts are listed in the accompanying audit report of 2 November 2026.

Additions made in this thesis

Two constants and one exact spectral identity appear in Chapter 7 and Section 6.4 and do not occur in the preprint [30]; their provenance is recorded here for completeness.

Uniquely determined. (v) The equal-blow-up coefficient $(\sqrt{5}+\sqrt{6}+3)/6$ of Proposition 7.1 is forced: it is $\Lambda_3(K_1 \vee C_5)+3$ divided by 6, and both $\Lambda_3(K_1 \vee C_5) = \sqrt{5}+\sqrt{6}$ and the shift $+3$ from the clique parts follow from Lemma 2.1 without any choice. (vi) The optimal rim weight $b^* = (7+5\sqrt{7})/126$ is the critical point of the one-variable quotient function computed in Lemma 3.2, and the apex weight $p^* = (91-25\sqrt{7})/126$ equals $1-5b^*$; neither is chosen.

Chosen, then verified. (iv) The orders $n \le 7$ for the exhaustive search, the blow-up sizes $q \in \{1,2,3,5,10,20,50,100,250,500\}$, and the weighted sample sizes 4000 per order in $n \in \{4,6,9,15,30,60\}$. These are sampling decisions; they affect the strength of the numerical evidence and nothing else.

Verification protocol for the additions. The exhaustive searches enumerated all $2^{n(n-1)/2}$ labelled graphs, computed $\lambda_1+\lambda_2+\lambda_3$ with a symmetric eigensolver in double precision, and compared against $cn-3$ evaluated from the exact expression for c; the margins observed are at least 2.2×10^{-2}, which is seven orders of magnitude above the double-precision error of a 7×7 symmetric eigenvalue computation. The number of labelled maximisers found at $n=6$ and $n=7$ is $72 = 6!/10$ and $252 = 7!/20$, matching $|\operatorname{Aut}(K_1 \vee C_5)| = 10$ and $|\operatorname{Aut}(K_2 \vee C_5)| = 2\cdot10 = 20$, which is the consistency check for uniqueness up to isomorphism. The exact identities $(\sqrt{5}+\sqrt{6}+3)/6 \cdot 6q - 3 = \Lambda_3(H[q])$ were re-verified in exact arithmetic in $\mathbb{Q}(\sqrt{5},\sqrt{6})$ for $q \le 10$ and in double precision for all q in the table.

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